

Fate Contagion and the Juggler Conjecture as an Almost-All Statement

Philippe Cochin
philippe@cochin.fr

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Abstract

The Juggler map is the integer map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even,} \\ \lfloor n^{3/2} \rfloor, & n \text{ odd.} \end{cases}$$

Every orbit has exactly one of three fates: it reaches 1, it enters a nontrivial cycle, or it is unbounded. It is conjectured that the first fate is the only one. This paper does not prove that conjecture. It proves that the fates are *contagious*: whichever fate occurs at all occurs on a set whose logarithmic counting function is at least a fixed power of $\log x$. Precisely, every nonempty set A of positive integers that is closed under taking preimages ($J(n) \in A \Rightarrow n \in A$) satisfies $\sum_{n \in A, n \leq x} 1/n \geq c(\log x)^\lambda$ for every $\lambda < \lambda^{**} = 0.4050 \dots$, whereas $\sum_{n \leq x} 1/n \sim \log x$. The mechanism is that the one-step preimages of the Juggler map are large and structured: the even preimages of m fill the interval $[m^2, (m+1)^2)$, and the two-step preimages through an even middle state fill a positive proportion of the fiber $\lfloor n^{3/4} \rfloor = m$, on which the parity of $\lfloor n^{3/2} \rfloor$ sweeps. The proof is elementary at this depth; it needs positive proportions on fibers, not equidistribution.

Two consequences follow. First, the Juggler conjecture is equivalent to a Tao-type almost-all statement with a *bounded* target and a logarithmic rate: every positive integer reaches 1 if and only if all but $O(y(\log y)^{-e})$ odd starts in $(y, 2y]$ enter the certified interval $[1, N_0]$, for some $e > 1 - \lambda^{**}$. For the Collatz map the corresponding implication is unavailable, because Collatz preimage trees can be thin; for the Juggler map descent is by powers and every preimage tree is contagious. Second, that almost-all statement follows from parity control on itinerary cylinders of depth $C \log_2 \log y$, in a hierarchy of forms whose weakest is a single exponential moment of the odd count on live starts; we identify what that weakest form does and does not need, show that no bounded-depth statement can reach it, and give a quantitative depth-uniformity budget for any analytic attack. An exact first-letter decomposition of the failure set shows that its single free term is the infinite-depth live mass controlled by the same hypothesis, so the termination problem has one frontier statement. The exact combinatorial layer is formalized in Lean 4; the counting is a human proof; numerical censuses are reported as observations only. No fate is excluded.

1. Introduction

The Juggler sequence was introduced by Pickover [1] and is sequence A094683 of the OEIS [2]. Starting from any positive integer, the orbit under J appears always to reach 1; this has been verified for all starts up to $3.5 \cdot 10^8$ [11]. The map belongs to the family of floor-power maps, and its dynamics resembles the $3x + 1$ problem [3, 4] in one respect and differs from it in another. The resemblance: a parity decides whether the orbit goes up or down, and the heuristic model of the orbit is a random walk with negative drift, here on the scale $\log \log n$ with steps $+\log_2 3 - 1$ (odd) and -1 (even). The difference: the Juggler map moves by *powers*, so a single contracting certificate sends n to n^e with $e < 1$, and the one-step preimages of an integer are not a residue class but an interval.

This paper is about the second feature and what it forces. Let R be the set of starts that reach 1, F its complement, $B(C)$ the basin of a nontrivial cycle C , and D the set of divergent starts. Each of these is *backward-closed*: if $J(n)$ belongs to it, so does n . We prove that every nonempty backward-closed set is large in logarithmic density, with an explicit exponent, by a downward recursion over two exact productions. Applied to the fate classes this says that no fate is rare: if a single start fails to reach 1, the failures have logarithmic count $\gg (\log x)^{0.40}$ and, on infinitely many dyadic blocks, natural density $\gg (\log y)^{-0.6}$. We call this *fate contagion*. It excludes no fate; it fixes the quantitative shape of the trichotomy.

The consequence for the conjecture is a change of form. Tao [6] proved that almost all Collatz orbits (in logarithmic density) attain almost bounded values, with a target $f(N) \rightarrow \infty$ arbitrarily slowly; a bounded target is out of reach there, and would not imply the Collatz conjecture even if available, because a hypothetical failure's preimage tree is only known to have $\geq x^{0.84}$ elements below x (Krasikov–Lagarias [7]). For the Juggler map contagion turns the implication into a theorem: a Tao-type statement with the *bounded* target $[1, N_0]$ and any logarithmic rate above $1 - \lambda^{**}$ is equivalent to the conjecture. We then show what such a statement needs. Because descent is by powers, the exponent walk has to travel only $L(y) = \log_2(\log 2y / \log N_0)$ units to bring a start of size y below N_0 ; a Chernoff count of the itineraries whose walk fails to do so within $C L(y)$ steps gives the rate, provided the itinerary cylinders of that depth are not over-populated. The depth is $O(\log \log y)$, where cylinders are large; for Collatz the analogous depth is $\Theta(\log y)$, where cylinders have single elements. This is the Juggler–Collatz inversion: for Collatz the Terras step (exact parity vectors) is free and renewal is the wall; for Juggler renewal is free and the Terras step — parity equidistribution of nested floor powers at unbounded depth — is the wall.

The last part of the paper locates that wall. The hypothesis has a hierarchy of forms; the weakest is one exponential moment of the odd count on the starts still above the floor (the live pressure), or its no-momentum form: odd-heavy prefixes do not bias the next parity toward odd, on average over depths. This form is insensitive to any $o(\log \log y)$ initial depths and to the all-odd tower cylinders, and no bounded-depth cylinder statement can imply it; an analytic method can afford to lose at most a factor $2^{1/C} \approx 1.04$ of its saving exponent per depth. An exact first-letter decomposition of the failure set has a single free term, and that term is the infinite-depth live mass of the OO cylinder — the same quantity. The termination problem therefore has one frontier statement, stated in Section 12.

1.0 Main results

Write $\ell_A(u, v] = \sum_{n \in A, u < n \leq v} 1/n$.

Theorem 1 (fate contagion). Let $A \subseteq \mathbb{N}$ be nonempty and backward-closed, and let $\lambda < \lambda^{**} = 0.4050 \dots$, the root of $2^{-\lambda} + \frac{5}{21}(\frac{3}{8})^\lambda + \frac{2}{21}(\frac{3}{4})^\lambda = 1$. There are $c > 0$ and x_0 , depending on A and λ , with

$$\sum_{\substack{n \in A \\ n \leq x}} \frac{1}{n} \geq c(\log x)^\lambda \quad (x \geq x_0).$$

Consequently each fate class that is realized by at least one start — R , the basin of any existing nontrivial cycle, the divergent set — satisfies this bound, and on infinitely many dyadic blocks has natural density $\gg (\log y)^{\lambda-1}$. (Theorem 5.2, Corollaries 5.3, 5.4.)

Theorem 2 (the conjecture as an almost-all statement). Fix N_0 with $[1, N_0] \subseteq R$. The following are equivalent: (i) every positive integer reaches 1; (ii) for some $\lambda < \lambda^{**}$, the starts $n \leq x$ whose orbit never enters $[1, N_0]$ have logarithmic count $o((\log x)^\lambda)$; (iii) for some $e > 1 - \lambda^{**} = 0.5950 \dots$ and all large y , $\#\{n \text{ odd} \in (y, 2y] : n \notin R\} \leq y(\log y)^{-e}$. (Corollary 5.5, Theorem 7.4, Theorem 7.5.)

Theorem 3 (Tao-type reduction). Let $L(y) = \log_2(\log 2y / \log N_0)$ and $d(y) = \lceil CL(y) \rceil$. Each of the following hypotheses implies (iii) of Theorem 2 with an explicit $e = e(C)$, hence the conjecture for $C \geq 21$: (a) no O -rooted itinerary cylinder of depth $d(y)$ exceeds its fair share $2^{-(d-1)}y/2$ among odd starts by more than $y(\log y)^{-A}$, $A > C + e(C)$ (Theorem 7.3); (b) one-sided: every cylinder of depth $1 \leq t < d(y)$ sends at most a fraction $q < \log 2 / \log 3$ of its members (plus $y(\log y)^{-A}$) to an odd next state (Theorem 8.1; $C \geq 46$ at $q = 0.55$); (c) the live pressure $\frac{1}{N} \sum_{n \text{ odd}, \tau(n) > d} e^{\theta o_d(n)} \leq (\frac{1}{2}(1 + e^\theta))^d e^{o(d)}$ at $\theta = \log(p_C / (1 - p_C))$, or

its no-momentum form $\sum_{t < d} (s_\theta(t) - q)^+ = o(d)$ for the tilted odd share s_θ of live starts (Theorem 8.2, Proposition 8.3). The hypotheses are not proved here. Any $o(\log \log y)$ initial depths are irrelevant to (c), tower cylinders biased below odd share 0.836 are harmless, and no cylinder statement of bounded depth implies any of (a)–(c) (Section 8.3).

Theorem 4 (odd generation and the exact map). A set that is both forward- and backward-closed and avoids 1 is the E -forest over the odd preimages of its intersection with $S = \{\lfloor m^{3/2} \rfloor : m \text{ odd}\}$; the conjecture holds iff no odd image fails. The logarithmic density of such a set satisfies the exact first-letter identity (6.1), whose single free term is the share of failures among OO -type odd starts (Theorem 6.1, Section 6.2).

Theorem 5 (one frontier statement). The free term of (6.1) is the infinite-depth live mass of the depth-two cylinder OO , bounded by every hypothesis of Theorem 3; the two halves of (6.1) are the contagion recursion and the failure-density upper recursion, with the same critical exponent $1 - \lambda_{\text{ideal}} = 0.5073$ as the rate threshold of Theorem 2 up to the fiber-constant gap; and an analytic cylinder bound can feed Theorem 3 at depth $C \log_2 \log y$ only if its saving exponent loses less than a factor $2^{1/C}$ per depth (Section 9).

Under a localization of Paper B’s triple parity discrepancy to sub-dyadic intervals (Section 10; Paper B is a working draft, so this part is conditional) the exponent improves to $\lambda^{***} = 0.4922 \dots$, the rate threshold to 0.5078, and the least depth constant to $C \geq 19$.

1.1 The three fates as currently constrained

The three fate classes are the three Moirai: Atropos cuts the thread (the class R), Lachesis measures an allotted length (a cycle basin), Clotho spins without end (divergence). What is known about each, all from outside this paper:

- *Atropos*. Every $n \leq N_0 = 3.5 \cdot 10^8$ reaches 1 (independently certified computation, [11, Section 5]); the Lean-verified floor is $N_0 = 260$. Paper B [12] proves that $7/8$ of all starts descend below themselves within five steps, with a power saving on dyadic blocks.
- *Lachesis*. A nontrivial cycle has minimum above $3.5 \cdot 10^8$ and period at least 780239 [11]; it has at least four even steps; its odd share is $\log 2 / \log 3$ up to $O(\Lambda/L)$, the critical share at which the exponent walk has zero drift.
- *Clotho*. A divergent orbit has unbounded exponent walk; its record jumps are quantized to the $\log_2 3$ lattice [11, laboratory extracts]. Nothing excludes it.

Fate contagion adds one sentence to each: whichever of the three acts at all acts on a set of logarithmic count $\gg (\log x)^{0.40}$.

1.2 Related work

Collatz parity vectors. Terras [4] and Everett [5] showed that the first k parities of the Collatz orbit of n are determined by $n \bmod 2^k$ and equidistributed exactly, and deduced that almost all starts have finite stopping time (descend below themselves). The analogous exact statement fails for the Juggler map: the parities of $\lfloor n^{3/2} \rfloor$, $\lfloor \lfloor n^{3/2} \rfloor^{3/2} \rfloor$, ... are Diophantine, not algebraic, and their equidistribution is proved only to depth four (Paper B, [12]).

Tao’s theorem. Tao [6] proved that for any $f(N) \rightarrow \infty$, almost all Collatz orbits (in logarithmic density) attain a value below $f(N)$. The bounded-target version is not available, and the renewal machinery through the Syracuse random variables is the heart of the proof. Section 7 explains why, for the Juggler map, the renewal step is unnecessary and the Terras step is the whole difficulty.

Preimage trees. Krasikov and Lagarias [7] proved that the Collatz preimage tree of any integer has $\geq x^{0.84}$ elements below x ; this is a lower bound of density zero, and nothing forces more. For the Juggler map the even preimages of m are an interval of length $2m + 1$, which is why backward-closed sets are contagious.

Floor-power maps and exponential sums. The parity of $\lfloor n^{3/2} \rfloor$ along an arithmetic progression is a question about $\{n^{3/2}/2\}$; Section 4 uses only an elementary sweep argument and, on even blocks, the second-derivative test with Vaaler’s approximation [10], as in Piatetski-Shapiro sequences. Deeper nesting is Paper B’s subject.

This laboratory. Paper A [11] proves period lower bounds for a hypothetical cycle from a certified descent floor (cycle financing, walk-charge envelope); Paper B [12] proves parity equidistribution of nested floor powers to depth four with power savings. Neither is reproved here; both enter as citations and, in Section 10, as a conditional strengthening.

1.3 Verification

Statements carry one of four roles. *Lean*: the exact combinatorial layer — closure of the fate classes, the trichotomy and the mutual exclusion of fates, the even block and the *OE* fiber as exact intervals, the fourth-power cell identity, odd generation, and the envelope descent into the floor — is formalized in Lean 4 (formal/Problems/Juggler/FateContagion.lean, no sorry; names in Appendix A). *Human proof*: the sweep lemma, the block average, the recursion and Theorem 1, odd generation’s density identity, the Chernoff, Azuma and exponential-moment arguments of Sections 7–9. *Verified computation*: the descent floor $3.5 \cdot 10^8$ and the period bound are Paper A’s; the closure of $[1, 260]$ under the two productions up to 10^9 is an exact computation. *Observation*: the fiber, block, Tao and pressure censuses of Section 11 are numerical checks of hypotheses and sanity checks of constants; they prove nothing and are labelled as such wherever they appear.

2. Fate classes and closure

$\mathbb{N} = \{1, 2, \dots\}$. For $A \subseteq \mathbb{N}$ and $0 \leq u < v$ write

$$\ell_A(u, v] = \sum_{\substack{n \in A \\ u < n \leq v}} \frac{1}{n}, \quad g_A(t) = \ell_A(e^{t/2}, e^t].$$

For $A = \mathbb{N}$, $g(t) = t/2 + O(e^{-t/2})$.

Definition. A is *backward-closed* if $J(n) \in A$ implies $n \in A$, and *forward-closed* if $n \in A$ implies $J(n) \in A$.

Lemma 2.1 (fate classes). Let $R = \{n : \exists k, J^k(n) = 1\}$, $F = \mathbb{N} \setminus R$, $B(m) = \{n : \exists k, J^k(n) = m\}$, and $D = \{n : \forall B \exists k, J^k(n) > B\}$. Each of $R, F, B(m), D$ is backward-closed; R, F, D are also forward-closed. Every $n \geq 1$ lies in R , or in $B(m)$ for some $m \geq 2$ on a cycle, or in D , and these possibilities exclude one another.

Proof. Backward closure is the definition of each set (for D : $J^k(J(n)) = J^{k+1}(n)$). Forward closure of R : if $J^k(n) = 1$ then $J^{k-1}(J(n)) = 1$ for $k \geq 1$, and $J(1) = 1$. Forward closure of D : given B , pick k with $J^k(n) > \max(B, n)$; then $k \geq 1$ and $J^{k-1}(J(n)) > B$. Trichotomy: a bounded orbit repeats, a repeat is a cycle, a cycle through 1 is the fate R . Exclusion: an orbit that reaches 1 or enters a cycle is bounded; an orbit that reaches 1 and passes through a cycle state $m \geq 2$ of period L would force $m = J^{qL}(m) = 1$. Lean: reachesOne_backwardClosed, not_reachesOne_backwardClosed, ancestor_backwardClosed, escapes_backwardClosed, reachesOne_floorPower, escapes_floorPower, fate_trichotomy, reachesOne_not_escapes, cycle_basin_not_escapes, reachesOne_not_cycle_basin. $\square$

Throughout Sections 3–5, A is a nonempty backward-closed set. Every statement applies to each fate class realized by at least one start.

3. The two productions

Lemma 3.1 (even block). Let $m \in A$. Every even n with $m^2 \leq n < (m+1)^2$ lies in A . Writing $E(m)$ for this set, $|E(m)| \geq m$ and

$$\sum_{n \in E(m)} \frac{1}{n} \geq \frac{m}{(m+1)^2} \geq \frac{1}{m} \left(1 - \frac{2}{m}\right), \quad \sum_{n \in E(m)} \frac{1}{n} \leq \frac{m+1}{m^2}.$$

Proof. $J(n) = \lfloor \sqrt{n} \rfloor = m$ exactly on that interval, so $n \in A$. The interval has $2m + 1$ integers, at least m and at most $m + 1$ of them even, each between m^2 and $(m + 1)^2$. Lean: `even_block_mem`, `even_block_card`. $\square$

Lemma 3.2 (OE fiber). Let $m \in A$ and

$$\Phi(m) = \{n \text{ odd} : m^4 \leq n^3 < (m + 1)^4\} = \{n \text{ odd} : m^{4/3} \leq n < (m + 1)^{4/3}\}.$$

If $n \in \Phi(m)$ and $\lfloor n^{3/2} \rfloor$ is even, then $J(J(n)) = m$, hence $n \in A$. Moreover $H_m := |\Phi(m)| \in [\frac{2}{3}m^{1/3} - 1, \frac{2}{3}(m + 1)^{1/3} + 1]$, and the fibers of distinct m are disjoint.

Proof. $J(n) = \lfloor \sqrt{n^3} \rfloor =: k$; if k is even then $J(k) = \lfloor \sqrt{k} \rfloor$, and $\lfloor \sqrt{\lfloor \sqrt{N} \rfloor} \rfloor = m \iff m^4 \leq N < (m + 1)^4$: this is the exact form of the transparent nesting $\lfloor \sqrt{\lfloor n^{3/2} \rfloor} \rfloor = \lfloor n^{3/4} \rfloor$. The interval $[m^{4/3}, (m + 1)^{4/3})$ has length between $\frac{4}{3}m^{1/3}$ and $\frac{4}{3}(m + 1)^{1/3}$ by the mean value theorem, and a half-open interval of length ℓ contains between $\ell/2 - 1$ and $\ell/2 + 1$ odd integers. Disjointness is the cell identity. Lean: `sqrt_sqrt_eq_iff`, `oe_fiber_mem`, `floorPower_oe_fiber`, `oe_fiber_disjoint`. $\square$

Write $G_m = \#\{n \in \Phi(m) : \lfloor n^{3/2} \rfloor \text{ even}\}$. The whole argument rests on lower bounds for G_m : an elementary bound on every fiber outside a thin exceptional set (Section 4.1), and a classical average over an even block of m 's (Section 4.2).

4. Parity on a fiber

4.1 The sweep lemma

On $\Phi(m)$ the quantity $x = n^{3/2}/2$ has $\lfloor n^{3/2} \rfloor$ even iff $\{x\} < \frac{1}{2}$. Consecutive odd n advance x by $\approx \frac{3}{2}n^{1/2} \approx \frac{3}{2}m^{2/3}$, and over the whole fiber this step drifts by less than one unit of $1/H_m$. So modulo 1 the fiber is an arithmetic progression with a nearly constant step α , of length H_m . Such a progression cannot avoid a half-circle unless its step is within $O(1/H_m)$ of 0, or its two-step is $(\alpha \approx \frac{1}{2})$ with the wrong phase).

Lemma 4.1 (sweep). Let $x_1 < x_2 < \dots < x_H$ be real numbers with $x_{j+1} - x_j \in [a, b]$ for all j , where $0 < a \leq b \leq \frac{1}{2}$, $b \leq \frac{21}{20}a$ and $(H - 1)a \geq 12$. Then

$$\#\{j : \{x_j\} < \tfrac{1}{2}\} \geq \tfrac{H}{7} \quad \text{and} \quad \#\{j : \{x_j\} \geq \tfrac{1}{2}\} \geq \tfrac{H}{7}.$$

The same holds with the cells $(k/2, (k + 1)/2]$ in place of $[k/2, (k + 1)/2)$, i.e. with the representative in $(0, 1]$ in place of the fractional part.

Proof. Cut $\mathbb{R}$ into cells $C_k = [k/2, (k + 1)/2)$, $k \in \mathbb{Z}$; even k are the *good* cells ($\{x\} < \frac{1}{2}$), odd k the *bad* cells. Call C_k *traversed* if $x_1 \leq k/2$ and $(k + 1)/2 \leq x_H$.

- (i) *A traversed cell contains at least $g := \lfloor 1/(2b) \rfloor \geq 1$ and at most $G := \lfloor 1/(2a) \rfloor + 1$ of the points.* Let j_0 be the least index with $x_{j_0} \geq k/2$. Then $x_{j_0} < k/2 + b$ (either $j_0 = 1$ and $x_1 = k/2$, or $x_{j_0-1} < k/2$). Since every step is at most b , $x_{j_0+i} < k/2 + (i + 1)b \leq (k + 1)/2$ for all $i \leq \lfloor 1/(2b) \rfloor - 1$, and these indices exist because $(k + 1)/2 \leq x_H$. That gives at least $\lfloor 1/(2b) \rfloor$ points in C_k . Since every step is at least a , the points in C_k are x_{j_0+i} with $k/2 + ia \leq x_{j_0+i} < (k + 1)/2$, so $i < 1/(2a)$ and there are at most $\lfloor 1/(2a) \rfloor + 1$ of them. The same count holds for any cell as an upper bound.
- (ii) *Counting cells.* The traversed cells are consecutive, and their number T is at least $2(x_H - x_1) - 2 \geq 2(H - 1)a - 2 \geq 22$. Good and bad traversed cells alternate, so their numbers T_g, T_b differ by at most one and $T_g, T_b \geq 10$. Every point outside the traversed cells lies in the cell of x_1 or the cell of x_H .
- (iii) *The ratio.* Hence $\text{Good} \geq gT_g$ and $H \leq G(T_g + T_b + 2) \leq G(2T_g + 3)$, so

$$\frac{\text{Good}}{H} \geq \frac{g}{G} \cdot \frac{T_g}{2T_g + 3} \geq \frac{g}{G} \cdot \frac{10}{23}.$$

Put $X = 1/(2a) \geq 1$, so $G = \lfloor X \rfloor + 1$ and, from $b \leq \frac{21}{20}a$, $g \geq \lfloor \frac{20}{21}X \rfloor$. If $X < 2$: $g \geq 1, G \leq 2$. If $2 \leq X < 3$: $g \geq 1, G = 3$. If $3 \leq X < 4$: $g \geq 2, G = 4$. If $4 \leq X < 5$: $g \geq 3, G = 5$. If $X \geq 5$:

$g/G \geq (\frac{20}{21}X - 1)/(X + 1) \geq 0.62$. In all cases $g/G \geq \frac{1}{3}$, so $\text{Good}/H \geq \frac{10}{69} > \frac{1}{7}$. The bad count is symmetric. For left-open cells the same proof applies verbatim (the first point in a cell $(c, c + \frac{1}{2}]$ after a point $\leq c$ is $\leq c + b \leq c + \frac{1}{2}$). $\square$

Lemma 4.2 (fiber parity). For $m \geq 10^6$ put $\alpha_m = \{\frac{3}{2}m^{2/3}\}$ and call m *good* if

$$\|\alpha_m\| \geq 22m^{-1/3} \quad \text{and} \quad \|\alpha_m - \tfrac{1}{2}\| \geq 2m^{-1/3},$$

where $\|\cdot\|$ is the distance to the nearest integer. If m is good then $G_m \geq \frac{1}{7}H_m$ and $H_m - G_m \geq \frac{1}{7}H_m$.

Proof. Let $n_1 < \dots < n_H$ be the odd integers of $\Phi(m)$, $n_{j+1} = n_j + 2$, and $x_j = n_j^{3/2}/2$. Then $\lfloor n_j^{3/2} \rfloor$ is even iff $\{x_j\} < \frac{1}{2}$. The steps are

$$\delta_j = x_{j+1} - x_j = \int_{n_j}^{n_{j+1}} \frac{3}{4}t^{1/2} dt \in [\frac{3}{2}n_j^{1/2}, \frac{3}{2}(n_j + 2)^{1/2}] \subseteq [A_m, B_m],$$

with $A_m = \frac{3}{2}m^{2/3}$ and $B_m = \frac{3}{2}((m+1)^{4/3} + 2)^{1/2}$. Using $\sqrt{u} - \sqrt{v} \leq (u-v)/(2\sqrt{v})$ and $(m+1)^{4/3} - m^{4/3} \leq \frac{4}{3}(m+1)^{1/3}$,

$$\eta_m := B_m - A_m \leq \frac{(m+1)^{1/3} + \frac{3}{2}}{m^{2/3}} \leq m^{-1/3} \left(1 + \frac{1}{3m} + \frac{3}{2}m^{-1/3}\right) \leq 1.02m^{-1/3}.$$

Also $H_m - 1 \geq \frac{2}{3}m^{1/3} - 2 \geq 0.646m^{1/3}$.

Case 1: $\alpha_m \leq \frac{1}{2} - 2m^{-1/3}$. Put $N = \lfloor A_m \rfloor$ and $y_j = x_j - jN$; then $\{y_j\} = \{x_j\}$ and the steps of y lie in $[a, b] = [\alpha_m, \alpha_m + \eta_m] \subseteq [22m^{-1/3}, \frac{1}{2}]$. Now $b/a \leq 1 + 1.02/22 < \frac{21}{20}$ and $(H-1)a \geq 0.646 \cdot 22 > 12$. Lemma 4.1 applies.

Case 2: $\alpha_m \geq \frac{1}{2} + 2m^{-1/3}$. Put $z_j = j(N+1) - x_j$, increasing with steps $(N+1) - \delta_j \in [1 - \alpha_m - \eta_m, 1 - \alpha_m] \subseteq [20.98m^{-1/3}, \frac{1}{2}]$; again $b/a \leq 1 + 1.02/20.98 < \frac{21}{20}$ and $(H-1)a \geq 0.646 \cdot 20.98 > 12$. Write $\langle z \rangle \in (0, 1]$ for the representative of z modulo 1 in $(0, 1]$. Then $\langle z_j \rangle = 1 - \{x_j\}$ for every j , so $\{x_j\} < \frac{1}{2} \iff \langle z_j \rangle \in (\frac{1}{2}, 1]$. Lemma 4.1 in its left-open form gives both counts.

The middle range $|\alpha_m - \frac{1}{2}| < 2m^{-1/3}$ and the range $\|\alpha_m\| < 22m^{-1/3}$ are excluded by goodness. $\square$

Lemma 4.3 (bad fibers are thin). For $u \geq 10^6$,

$$\#\{m \in (u, 2u] : m \text{ bad}\} \leq 63u^{2/3}, \quad \sum_{\substack{m > U \\ m \text{ bad}}} \frac{1}{m} \leq 306U^{-1/3} \quad (U \geq 10^6).$$

Proof. $\varphi(m) = \frac{3}{2}m^{2/3}$ increases with $\varphi(m+1) - \varphi(m) \in [(m+1)^{-1/3}, m^{-1/3}] \subseteq [(3u)^{-1/3}, u^{-1/3}]$ on $(u, 2u]$. Badness means $\{\varphi(m)\}$ lies in one of two arcs of the circle, of total length at most $48u^{-1/3}$. As m runs over $(u, 2u]$, φ increases by $\frac{3}{2}u^{2/3}(2^{2/3} - 1) < 0.882u^{2/3}$, so it passes each arc at most $0.882u^{2/3} + 2$ times, and each pass through an arc of length w takes at most $w(3u)^{1/3} + 1$ values of m . Hence the count is at most $(0.882u^{2/3} + 2)(48 \cdot 3^{1/3} + 2) \leq 63u^{2/3}$ for $u \geq 10^6$. Summing $63(2^i U)^{-1/3}$ over $i \geq 0$ gives $63U^{-1/3}/(1 - 2^{-1/3}) \leq 306U^{-1/3}$. $\square$

4.2 The block average

The elementary bound $G_m \geq H_m/7$ is far from the truth: the census of Section 11 finds mean $G_m/H_m = \frac{1}{2}$ and a minimum near $\frac{1}{3}$ on good fibers. When the members of A come in even blocks — as the E -produced part of A always does — the fibers can be averaged over the block, and the average is exactly $\frac{1}{2}$ by a classical exponential-sum estimate.

Proposition 4.4 (block average). For $m' \geq 2$ let $I(m') = [m'^{8/3}, (m' + 1)^{8/3})$ and

$$U(m') = \{n \text{ odd in } I(m') : \lfloor n^{3/4} \rfloor \text{ even}, \lfloor n^{3/2} \rfloor \text{ even}\} = \bigsqcup_{m \in E(m')} \{n \in \Phi(m) : \lfloor n^{3/2} \rfloor \text{ even}\}.$$

There is an absolute constant C_0 with

$$|U(m')| \geq \frac{1}{4} \#\{n \text{ odd in } I(m')\} - C_0 m'^{11/9} \log(m' + 1).$$

Since $\#\{n \text{ odd in } I(m')\} \geq \frac{4}{3}m'^{5/3} - 1$, this reads $|U(m')| \geq \frac{1}{3}m'^{5/3}(1 - \varepsilon_B(m'))$ with $\varepsilon_B(m') = \frac{3}{4}m'^{-5/3} + 3C_0m'^{-4/9} \log(m' + 1) \rightarrow 0$.

Proof. Write $\psi(y) = (-1)^{\lfloor y \rfloor}$, so that $\lfloor \lfloor y \rfloor \text{ even} \rfloor = \frac{1}{2}(1 + \psi(y))$, and

$$|U(m')| = \frac{1}{4} \sum_n (1 + \psi(n^{3/4}) + \psi(n^{3/2}) + \psi(n^{3/4})\psi(n^{3/2})),$$

the sum over odd $n \in I(m')$. Four terms.

Term 1 is the main term.

Term 2. $\lfloor n^{3/4} \rfloor = m$ is constant on each fiber, so $\sum \psi(n^{3/4}) = \sum_{m'^2 \leq m < (m'+1)^2} (-1)^m H_m$. Since H_m is within 1 of half the fiber length and consecutive fiber lengths differ by at most 1, $|H_{m+1} - H_m| \leq 3$; pairing consecutive m gives $|\text{Term 2}| \leq 3m' + \max H_m \ll m'$.

Term 3. Expand ψ by Vaaler's theorem [10]: $\psi(y) = V_J(y/2) + O(\Delta_J(y/2))$, where $V_J(t) = \sum_{0 < |q| \leq J} a_q e(qt)$ with $|a_q| \ll 1/|q|$ and $\Delta_J \geq 0$ is a trigonometric polynomial of degree J with constant term and coefficients $\leq 1/(J+1)$. Reindex $n = 2r + 1$. For $q \neq 0$ the phase $f(r) = q(2r+1)^{3/2}/2$ has $f''(r) = \frac{3}{2}q n^{-1/2}$, which on $I(m')$ lies between $\lambda_q = \frac{3}{2}|q|(m'+1)^{-4/3}$ and $2^{4/3}\lambda_q$. The second-derivative test (van der Corput; [13, Theorem 2.2]) with interval length $M \leq \frac{4}{3}(m'+1)^{5/3} + 1$ gives

$$\left| \sum_r e(f(r)) \right| \ll M \lambda_q^{1/2} + \lambda_q^{-1/2} \ll |q|^{1/2} m' + |q|^{-1/2} m'^{2/3}.$$

Summing against $|a_q| \ll 1/|q|$ over $0 < |q| \leq J$ gives $\ll J^{1/2} m' + m'^{2/3}$. The Δ_J error contributes $\ll M/J + J^{-1} \sum_{0 < |q| \leq J} |\sum_r e(f)| \ll m'^{5/3}/J + J^{1/2} m'$. With $J = m'^{4/9}$ both are $\ll m'^{11/9}$.

Term 4. Expand both factors. The main part is $\sum_{q_1, q_2 \neq 0} a_{q_1} a_{q_2} \sum_r e(f_{q_1, q_2}(r))$ with $f_{q_1, q_2}(r) = \frac{1}{2}(q_1(2r+1)^{3/4} + q_2(2r+1)^{3/2})$ and

$$f''_{q_1, q_2}(r) = \frac{3}{2}q_2 n^{-1/2} \left(1 - \frac{q_1}{4q_2} n^{-3/4} \right).$$

For $|q_1| \leq J = m'^{4/9}$ and $n \geq m'^{8/3}$ the correction is $\leq \frac{1}{4}m'^{4/9-2}$, so $|f''|$ is again between λ_{q_2} and $2^{4/3}\lambda_{q_2}$ up to a factor $1 + o(1)$, uniformly in q_1 ; the second-derivative test gives the same bound as in Term 3, and the double sum against $|a_{q_1} a_{q_2}| \ll 1/(|q_1| |q_2|)$ is $\ll (\log J)(J^{1/2} m' + m'^{2/3})$. Three error terms remain. $\sum \Delta_J(n^{3/2}/2)$ is bounded as in Term 3. For $\sum \Delta_J(n^{3/4}/2)$ the constant term gives $M/(J+1)$ and each mode $0 < |q_1| \leq J$ gives $(J+1)^{-1} |\sum_r e(q_1(2r+1)^{3/4}/2)|$; here the phase has derivative $\frac{3}{4}q_1 n^{-1/4}$, monotone and of size between $\frac{3}{4}|q_1|(m'+1)^{-2/3}$ and $\frac{3}{4}m'^{4/9-2/3} < \frac{1}{2}$, so the Kusmin–Landau first-derivative test bounds the sum by $\ll m'^{2/3}/|q_1|$; the modes contribute $\ll m'^{2/3} \log J/J$ in total. The product error $\sum \Delta_J \Delta_J$ is at most $\sup \Delta_J \leq 2$ times either of the two previous sums. Altogether Term 4 is $\ll m'^{11/9} \log m'$.

Collecting, $|U(m')| = \frac{1}{4} \#\{n \text{ odd in } I(m')\} + O(m'^{11/9} \log(m' + 1))$. $\square$

The proposition is not sharp; it is all the recursion needs.

5. The recursion and the theorem

5.1 Three sources of members of A in $(\sqrt{x}, x]$

Let $x \geq 2$ and $t = \log x$. Three families of members of $A \cap (\sqrt{x}, x]$ are pairwise disjoint.

1. **E -images.** For $m \in A \cap (x^{1/4}, \sqrt{x} - 1]$, $E(m) \subseteq (\sqrt{x}, x]$ (as $m^2 > \sqrt{x}$ and $(m+1)^2 \leq x$). Distinct m give disjoint blocks. By Lemma 3.1 the log-mass is at least $\sum_m \frac{1}{m}(1 - \frac{2}{m}) \geq (1 - 2x^{-1/4}) \ell_A(x^{1/4}, \sqrt{x} - 1]$, and $\ell_A(x^{1/4}, \sqrt{x} - 1] \geq g_A(t/2) - 2x^{-1/2}$ (at most one integer lies in $(\sqrt{x} - 1, \sqrt{x}]$).
2. **OE -images of E -blocks.** For $m' \in A \cap (x^{3/16}, x^{3/8} - 1]$, $U(m') \subseteq A \cap (\sqrt{x}, x]$ (Lemma 3.2 applied to the even $m \in E(m') \subseteq A$; $I(m') \subseteq (\sqrt{x}, x]$ as $m'^{8/3} > \sqrt{x}$ and $(m' + 1)^{8/3} \leq x$). Distinct m' give disjoint $I(m')$. Each $n \in U(m')$ has $1/n > (m' + 1)^{-8/3}$, so by Proposition 4.4 the log-mass of $U(m')$ is at least $\frac{1}{3}m'^{5/3}(1 - \varepsilon_B(m'))(m' + 1)^{-8/3} \geq \frac{1}{3m'}(1 - \varepsilon_B(m') - \frac{8}{3m'})$, and summing, $\geq \frac{1}{3}(1 - \varepsilon'(x))(g_A(3t/8) - 2x^{-3/8})$ with $\varepsilon'(x) = \sup_{m' > x^{3/16}}(\varepsilon_B(m') + \frac{8}{3m'}) \rightarrow 0$.
3. **OE -images of the rest.** Let $A_x^E = \bigcup_{m'' \in A} E(m'') \cap (x^{3/8}, x^{3/4}]$ be the E -produced part of $A \cap (x^{3/8}, x^{3/4}]$ and A_x^{rest} its complement in $A \cap (x^{3/8}, x^{3/4}]$. The blocks meeting $(x^{3/8}, x^{3/4}]$ come from $m'' \in [x^{3/16} - 1, x^{3/8}]$, so $\ell(A_x^E) \leq \sum_{m''} \frac{m''+1}{m''^2} \leq (1 + 2x^{-3/16})(g_A(3t/8) + 2x^{-3/16})$ and $\ell(A_x^{\text{rest}}) \geq g_A(3t/4) - (1 + 2x^{-3/16})(g_A(3t/8) + 2x^{-3/16})$. For $m \in A_x^{\text{rest}}$ with $m \leq x^{3/4} - 1$ (this drops at most one element, of log-mass $\leq 2x^{-3/4}$), $\Phi(m) \subseteq (\sqrt{x}, x]$; the fibers are disjoint from each other and from those of item 2 (different m), and for good $m \geq 10^6$ Lemmas 4.2 and 3.2 give log-mass at least $\frac{1}{7}(\frac{2}{3}m^{1/3} - 1)(m+1)^{-4/3} \geq \frac{2}{21m}(1 - 2m^{-1/3})$. Bad $m > x^{3/8}$ carry log-mass at most $306x^{-1/8}$ (Lemma 4.3). Hence item 3 contributes at least $\frac{2}{21}(1 - 2x^{-1/8})[\ell(A_x^{\text{rest}}) - 306x^{-1/8} - 2x^{-3/4}]_+$.

The images in item 1 are even, those in items 2–3 odd, so the three families are disjoint, and all lie in $(\sqrt{x}, x]$.

5.2 The functional inequalities

Adding items 1 and 2:

$$g_A(t) \geq (1 - \varepsilon_1(t))g_A(t/2) + \frac{1}{3}(1 - \varepsilon_2(t))g_A(3t/8) - \varepsilon_3(t), \quad (5.1)$$

with $\varepsilon_1 = 2e^{-t/4}$, $\varepsilon_2 = \varepsilon'(e^t)$, $\varepsilon_3 = 2e^{-t/2} + e^{-3t/8}$, all tending to 0. Adding item 3 as well, and noting that the coefficient of the *actual* value $g_A(3t/8)$ is $\frac{1}{3}(1 - \varepsilon_2) - \frac{2}{21}(1 + 2e^{-3t/16}) \rightarrow \frac{5}{21} > 0$:

$$g_A(t) \geq (1 - \varepsilon_1)g_A(t/2) + (\frac{5}{21} - \varepsilon_4)g_A(3t/8) + (\frac{2}{21} - \varepsilon_5)g_A(3t/4) - \varepsilon_6(t), \quad (5.2)$$

with $\varepsilon_4, \varepsilon_5, \varepsilon_6 \rightarrow 0$ (for t so large that $x^{3/8} \geq 10^6$; the bracket $[\cdot]_+$ may be dropped because the right side of (5.2) is a lower bound for it). Every coefficient of a g_A -value on the right is nonnegative for large t , so lower bounds for g_A at $t/2$, $3t/8$, $3t/4$ transfer.

5.3 The seed

Lemma 5.1 (seed). Every nonempty backward-closed A contains an integer $m \geq 3$, and for any such m ,

$$g_A(t) \geq c_A := \left(1 - \frac{2}{m^4}\right) \left(\frac{0.375}{m+1} - \frac{1}{(m+1)^2 - 1}\right) > 0 \quad \text{for all } t \geq 4 \log(m+1).$$

Proof. If $A \ni a \leq 2$ then $E(a) \subseteq A$ contains 2 or 4, and $E(2) \ni 4$, so $A \ni 4$. Fix $m \geq 3$ in A and let $S_1 = E(m)$, $S_{k+1} = \bigcup_{m'' \in S_k} E(m'')$. Then $S_k \subseteq A \cap [m^{2^k}, (m+1)^{2^k}]$ and, by Lemma 3.1, $\ell(S_{k+1}) \geq (1 - 2m^{-2^k})\ell(S_k)$, so $\ell(S_k) \geq \ell(S_1) \prod_{j \geq 1} (1 - 2m^{-2^j}) \geq 0.75 \ell(S_1) \geq \frac{0.375}{m+1}$ for $m \geq 3$. Let $y \geq (m+1)^4$ and choose $k \geq 2$ with $(m+1)^{2^k} \leq y < (m+1)^{2^{k+1}}$. Then $S_k \subseteq (y^{1/4}, y]$ (as $y < m^{2^{k+2}}$). The members of S_k above $\sqrt{y}$ lie in $(\sqrt{y}, y]$; the members in $(y^{1/4}, \sqrt{y} - 1]$ have their E -images in $(\sqrt{y}, y]$, with log-mass at least $(1 - 2m^{-2^k})$ times their own; at most one member lies in $(\sqrt{y} - 1, \sqrt{y}]$, with $1/n \leq 1/((m+1)^2 - 1)$. Hence $g_A(\log y) \geq (1 - 2m^{-4})(\ell(S_k) - 1/((m+1)^2 - 1)) \geq c_A$, and $c_A > 0$ because $0.375(m+1) \geq 1 + 0.375/(m+1)$ for $m \geq 3$. $\square$

5.4 The theorem

Let λ^* and λ^{**} be the roots in $(0, 1)$ of

$$2^{-\lambda} + \frac{1}{3}\left(\frac{3}{8}\right)^\lambda = 1, \quad 2^{-\lambda} + \frac{5}{21}\left(\frac{3}{8}\right)^\lambda + \frac{2}{21}\left(\frac{3}{4}\right)^\lambda = 1.$$

Numerically $\lambda^* = 0.3774\dots$ and $\lambda^{**} = 0.4050\dots$. For comparison: the elementary sweep alone, without Proposition 4.4, gives $2^{-\lambda} + \frac{2}{21}\left(\frac{3}{4}\right)^\lambda = 1$, root $0.138\dots$; perfect fiber equidistribution at this depth would give $2^{-\lambda} + \frac{1}{3}\left(\frac{3}{4}\right)^\lambda = 1$, root $\lambda_{\text{ideal}} = 0.4927\dots$, the ceiling of the depth-two method.

Theorem 5.2 (logarithmic density of a backward-closed set). Let $A \subseteq \mathbb{N}$ be nonempty and backward-closed, and let $0 < \lambda < \lambda^{**}$. There are $c > 0$ and x_0 , depending on A and λ , such that

$$\sum_{\substack{n \in A \\ n \leq x}} \frac{1}{n} \geq c(\log x)^\lambda \quad \text{for all } x \geq x_0.$$

The same conclusion for $\lambda < \lambda^*$ uses only Proposition 4.4 (no sweep lemma).

Proof. Set $\zeta = 2^{-\lambda} + \frac{5}{21}\left(\frac{3}{8}\right)^\lambda + \frac{2}{21}\left(\frac{3}{4}\right)^\lambda - 1 > 0$. By Lemma 5.1 and the decay of the ε_i , choose t_1 with $\frac{3}{8}t_1 \geq 4\log(m+1)$ such that for all $t \geq t_1$

$$\varepsilon_1(t)2^{-\lambda} + \varepsilon_4(t)\left(\frac{3}{8}\right)^\lambda + \varepsilon_5(t)\left(\frac{3}{4}\right)^\lambda \leq \frac{\zeta}{3}, \quad \varepsilon_6(t) \leq \frac{2\zeta}{3}c_A,$$

and all coefficients in (5.2) are nonnegative. Put $K = c_A t_1^{-\lambda}$. Claim: $g_A(t) \geq Kt^\lambda$ for all $t \geq \frac{3}{8}t_1$. For $t \in [\frac{3}{8}t_1, t_1]$ this is Lemma 5.1 ($Kt^\lambda \leq c_A$). Suppose it holds on $[\frac{3}{8}t_1, T]$ with $T \geq t_1$ and let $t \in (T, \frac{8}{3}T]$. Then $t/2, 3t/8, 3t/4 \in [\frac{3}{8}t_1, T]$, so by (5.2)

$$g_A(t) \geq Kt^\lambda \left[(1 + \zeta) - \frac{\zeta}{3} \right] - \varepsilon_6(t) \geq Kt^\lambda \left(1 + \frac{2\zeta}{3} \right) - \frac{2\zeta}{3}c_A \geq Kt^\lambda,$$

using $Kt^\lambda \geq Kt_1^\lambda = c_A$. Induction on the intervals $(\frac{8}{3})^N T$ covers all $t \geq \frac{3}{8}t_1$. Finally $\sum_{n \in A, n \leq x} 1/n \geq g_A(\log x)$. For $\lambda < \lambda^*$ run the same argument with (5.1). $\square$

Corollary 5.3 (natural density, infinitely often). For every $\lambda < \lambda^{**}$ there is $c > 0$ such that for every $X \geq x_0$ some $y \in (\sqrt{X}, X]$ satisfies $\#(A \cap (y/2, y]) \geq c y (\log y)^{\lambda-1}$.

Proof. $\ell_A(\sqrt{X}, X] = g_A(\log X) \geq K(\log X)^\lambda$ is spread over at most $\frac{1}{2}\log_2 X + 2$ dyadic blocks $(y/2, y] \subseteq (\sqrt{X}, 2X]$; one of them has $\ell_A(y/2, y] \geq K'(\log X)^{\lambda-1}$, hence $\#(A \cap (y/2, y]) \geq \frac{y}{2} \ell_A(y/2, y] \geq \frac{K'}{2} y (2\log y)^{\lambda-1}$, because $\log X \leq 2\log y$ and $\lambda - 1 < 0$. $\square$

Corollary 5.4 (fate contagion). Let φ be a fate realized by at least one start: the trivial cycle, a particular nontrivial cycle C , or divergence. Then $\{n : \text{fate}(n) = \varphi\}$ satisfies Theorem 5.2 and Corollary 5.3. In particular:

1. $\sum_{n \in R, n \leq x} 1/n \gg (\log x)^\lambda$ for every $\lambda < \lambda^{**}$.
2. If some start does not reach 1, then $\sum_{n \notin R, n \leq x} 1/n \gg (\log x)^\lambda$, and on infinitely many dyadic blocks the failures have natural density $\gg (\log y)^{\lambda-1}$.
3. If a nontrivial cycle exists, the set of starts that enter it has the same lower bounds; if a divergent orbit exists, so does the set of divergent starts.

Proof. Lemma 2.1 and Theorem 5.2. $\square$

Corollary 5.5 (the conjecture as an almost-all statement, logarithmic form). Fix N_0 with $[1, N_0] \subseteq R$ (Lean: $N_0 = 260$; certified computation: $N_0 = 3.5 \cdot 10^8$). The following are equivalent.

1. Every $n \geq 1$ reaches 1.
2. For some $\lambda < \lambda^{**}$, $\sum_{n \leq x, n \notin R} 1/n = o((\log x)^\lambda)$.
3. For some $\lambda < \lambda^{**}$, the starts $n \leq x$ whose orbit never enters $[1, N_0]$ have logarithmic count $o((\log x)^\lambda)$.

Proof. (1) $\Rightarrow$ (3): the set is empty. (3) $\Rightarrow$ (2): an orbit that enters $[1, N_0]$ reaches 1, so F is contained in the set of (3). (2) $\Rightarrow$ (1): F is backward-closed; if it were nonempty, Theorem 5.2 would contradict (2). $\square$

In natural-density language, (3) asks that all but $O(x(\log x)^{-0.6})$ starts below x eventually fall below $3.5 \cdot 10^8$: a Tao-type “almost all orbits attain a bounded value” statement with a mild logarithmic rate. Section 7 states it for dyadic blocks of odd starts (Theorem 7.4) and shows what it follows from.

5.5 Remarks

Why the exponent is not 1. Positive logarithmic density ($\lambda = 1$) would need the productions to account for all of the mass: in the ideal recursion $g(t) \geq g(t/2) + \sum_w c_w g(e_w t)$ over descent certificates w with landing exponents e_w , linear growth requires the certificate classes to carry the full odd mass. Depth two (E, OE) carries $\frac{3}{4}$ of the integers and gives at best $\lambda_{\text{ideal}} = 0.4927$; Paper B’s depth-five classes ($\frac{7}{8}$) would give about 0.75; the OO -rooted classes send their mass to *larger* scales ($x^{3/2}$), which a downward induction cannot use until a later descent certificate returns it. Positive density of R is therefore as hard as the full almost-all-descent statement with a rate, and is not claimed.

Pointwise natural density. Theorem 5.2 is logarithmic and Corollary 5.3 holds infinitely often, not for every x . A pointwise bound $\#(A \cap [1, x]) \gg x(\log x)^{\lambda-1}$ cannot be inducted through fixed-ratio intervals — the E -preimage of a ratio- r interval has ratio $\sqrt{r}$ — and for a backward-closed set generated by a single seed it is false: the E -tree of one integer is lacunary at the dyadic scale. For R itself the E -tree of the full interval $[1, N_0]$ is uniform on dyadic blocks as long as $2^k \ll N_0$ at level k , i.e. for $\log x \ll N_0 \log N_0$; the density there is at least $2^{-k} \approx \log N_0 / (2 \log x)$ from E alone.

Sharpness of the constants. The fiber lemma gives $\frac{1}{7}$; the truth on good fibers is about $\frac{1}{3}$ (Section 11.1). Replacing $\frac{1}{7}$ by $\frac{1}{3}$ would move λ^{**} from 0.405 to 0.448; the ceiling of the depth-two method is 0.4927.

What is excluded. Nothing. Corollary 5.4 does not say that a cycle or a divergent orbit is impossible; it says that either would be common.

6. Odd generation and the exact first-letter decomposition

This section records what the contagion viewpoint says the termination problem *is*, exactly. Write $S = \{\lfloor m^{3/2} \rfloor : m \text{ odd}\}$ for the image of the odd integers, and S_{odd} for its odd members.

6.1 Odd generation

Theorem 6.1 (odd generation; Lean). Let A be forward- and backward-closed with $1 \notin A$. Then:

1. every $n \in A$ descends by even steps only to an odd member $m \geq 3$ of A (**exists_odd_ancestor_ge_three**);
2. for odd n , $n \in A \iff \lfloor n^{3/2} \rfloor \in A$ (**odd_mem_iff**), so the odd members of A are exactly the odd preimages of $A \cap S$;
3. $A \neq \emptyset \iff A \cap S \neq \emptyset$, i.e. A contains the image of some odd $m \geq 3$ (**nonempty_iff_odd_image_mem**).

Proof. (1) If $n \in A$ is even, $\lfloor \sqrt{n} \rfloor \in A$ by forward closure and is smaller; the chain stops at an odd integer, which is ≥ 3 since $1 \notin A$ and $2 \notin A$ ($J(2) = 1$). (2) Forward closure gives one direction, backward closure the other. (3) follows from (1) and (2). $\square$

Applied to the failure set F : every positive integer reaches 1 iff no image $\lfloor m^{3/2} \rfloor$ of an odd m fails to reach 1, and F is the E -forest over the odd preimages of $F \cap S$. The whole class is generated by the sparse set $F \cap S$ (the image set S has density $\asymp z^{-1/3}$ at scale z).

6.2 The exact decomposition and the free term

Let A be forward- and backward-closed. On $(\sqrt{x}, x]$ its members split by first letters into three exact pieces:

- the **even** members, $= \bigsqcup_{m \in A} E(m) \cap (\sqrt{x}, x]$, of log-mass $\ell_A(x^{1/4}, \sqrt{x}) + O(x^{-1/4})$ (Lemma 3.1);

- the **OE-type** odd members ($\lfloor n^{3/2} \rfloor$ even), $= \bigsqcup_{m \in A} \{n \in \Phi(m) : \lfloor n^{3/2} \rfloor \text{ even}\}$ (Lemma 3.2), of log-mass between $\frac{2}{21}$ and $\frac{4}{7}$ of $\ell_A(x^{3/8}, x^{3/4}]$ on good fibers (Lemma 4.2, two-sided) and exactly $\frac{1}{3}(1 + o(1))$ of it on E -blocks (Proposition 4.4);
- the **OO-type** odd members ($\lfloor n^{3/2} \rfloor$ odd), $=$ the odd preimages of $A \cap S_{\text{odd}} \cap (x^{3/4}, x^{3/2}]$, of log-mass $\sum_{m \in A \cap S_{\text{odd}}} 1/n(m)$, where $n(m)$ is the unique odd preimage.

In log-density form, with $\varphi_A(t) = \ell_A(e^{t/2}, e^t]/(t/2)$ and $\psi_A(t)$ the log-weighted fraction of **OO-type** odd $n \in (\sqrt{x}, x]$ whose image lies in A :

$$\varphi_A(t) = \frac{1}{2}\varphi_A(t/2) + \frac{1}{4}\varphi_A^{\text{fib}}(3t/4) + \frac{1}{4}\psi_A(t) + O(e^{-t/4}/t), \quad (6.1)$$

where φ_A^{fib} is the fiber-weighted density (equal to φ_A on E -blocks). For $A = \mathbb{N}$ every term is 1. For $A = F$ the boundary condition is $\varphi_F(t) = 0$ for $t \leq \log N_0$. **The single free term is ψ_F :** if $\psi_F(t) \leq \varphi_F(3t/2)$ for all t — the failures are not over-represented among the odd images S_{odd} — then (6.1) is the harmonic inequality of a random walk on $\log t$ with steps $\log \frac{1}{2}$, $\log \frac{3}{4}$, $\log \frac{3}{2}$ of probabilities $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{4}$, whose drift -0.317 is negative, and the boundary condition forces $\varphi_F \equiv 0$. Conversely $\psi_F \equiv 1$ (every **OO-type** odd number above the floor fails) is consistent with (6.1). Hence:

The Juggler conjecture is equivalent to the S -fairness of the failure set: $F \neq \emptyset$ forces the failures to be over-represented among $\{\lfloor m^{3/2} \rfloor : m \text{ odd}\}$ at some scales.

Section 9 identifies ψ_F with a quantity controlled by the hypotheses of Sections 7–8 and shows that the two halves of (6.1) are, respectively, the contagion recursion of Section 5 and an upper recursion for the failure density with the same critical exponent.

Where Papers A and B sit. Paper A constrains the fate Lachesis from the inside: finance and the walk charge bound the *states* of a hypothetical cycle (minimum $> 3.5 \cdot 10^8$, period ≥ 780239). Contagion constrains the basin: if the cycle exists, its basin is a two-way closed class with the cycle's states as seeds and log-count $\gg (\log x)^{0.405}$. The two do not meet: finance bounds the seed, contagion the growth from the seed, and no inequality bounds a basin from above. Paper B controls the descending branches of (6.1) — the fairness of the landing distributions of E , OE and, with its depth-4 and depth-5 theorems, of deeper contracting words — on dyadic blocks; Section 10 uses one of them on sub-dyadic intervals. Neither touches ψ_F : the ascending branch OO sends mass to $x^{3/2}$, and its return is the nested-floor parity at all depths.

7. The Tao-type reduction

7.1 Notation

Fix $N_0 \geq 2$ with $[1, N_0] \subseteq R$ (Lean: $N_0 = 260$; certified computation: $N_0 = 3.5 \cdot 10^8$). For a word $w \in \{O, E\}^d$ and $t \leq d$ write $o_t(w)$ for the number of odd letters among the first t and

$$u_t(w) = o_t(w) \log_2 3 - t$$

for the *exponent walk*, so that the ideal image after t letters is $n^{2^{u_t}}$. For $y \geq 1$ put

$$L(y) = \log_2 \frac{\log 2y}{\log N_0}, \quad d(y) = \lceil C L(y) \rceil,$$

with a constant $C \geq 5$ fixed below. A word w of length d is L -bad if $u_t(w) > -L$ for every $1 \leq t \leq d$ (its walk never descends to $-L$). $\text{word}_d(n)$ is the realized itinerary of the first d steps of n (letter s is the parity of $J^{s-1}(n)$); the *cylinder* $[w]_y$ is the set of odd $n \in (y, 2y]$ with $\text{word}_{|w|}(n) = w$. Odd starts have first letter O , so the fair share of an O -rooted cylinder of depth d among the $N = y/2$ odd starts of $(y, 2y]$ is $2^{-(d-1)}N$.

7.2 Envelope descent

Lemma 7.1. Let $n \in (y, 2y]$ and $d \geq 1$. If $\text{word}_d(n)$ is not $L(y)$ -bad, then $n \in R$.

Proof. Some $t \leq d$ has $u_t \leq -L(y)$, i.e. $3^{o_t}/2^t \leq \log N_0 / \log 2y \leq \log N_0 / \log n$, i.e. $n^{3^{o_t}} \leq N_0^{2^t}$ as integers. The power envelope $J^t(n)^{2^t} \leq n^{3^{o_t}}$ (Paper A, Theorem 2.2; `power_bound_word`) gives $J^t(n) \leq N_0$, so

$J^t(n) \in R$ and $n \in R$ by backward closure. Lean: `iterate_le_of_envelope`, `mem_of_envelope_floor`, `reachesOne_of_itinerary_envelope`. $\square$

The descent is by powers: this is why the depth needed is $O(\log \log y)$, not $O(\log y)$.

7.3 Bad words are rare

Lemma 7.2 (Chernoff). For $C \geq 5$ put

$$p_C = \frac{1 - 1/C}{\log_2 3}, \quad e(C) = \frac{C D(p_C \| \frac{1}{2})}{\ln 2}, \quad D(p \| \frac{1}{2}) = p \ln(2p) + (1 - p) \ln(2(1 - p)).$$

For $L > 0$ and $d \geq CL$, the number of L -bad words of length d is at most $2^d \cdot 2^{-e(C)L}$. Moreover, for every $\varepsilon > 0$ there is L_0 such that for $L \geq L_0$ and $d = \lceil CL \rceil$ the number of L -bad words of length d beginning with O is at most $2^{d-1} \cdot 2^{-(e(C)-\varepsilon)L}$.

Proof. An L -bad word has $u_d > -L$, i.e. $o_d > (d - L)/\log_2 3 \geq p_C d$ (as $L/d \leq 1/C$). For a fair coin, $\Pr[\text{Bin}(d, \frac{1}{2}) \geq pd] \leq e^{-dD(p \| 1/2)}$ for $p \geq \frac{1}{2}$, and $D(\cdot \| \frac{1}{2})$ is increasing on $[\frac{1}{2}, 1]$; $p_C \geq \frac{1}{2}$ for $C \geq 5$. Hence the count is at most $2^d e^{-dD(p_C \| 1/2)} \leq 2^d e^{-CL D(p_C \| 1/2)} = 2^d 2^{-e(C)L}$. For the O -rooted count, the remaining $d - 1$ letters contain $o_d - 1 > (d - L)/\log_2 3 - 1 = p'(d - 1)$ odd letters with $p' = p_C - O(1/L)$; the same bound with $d - 1$ trials gives $2^{d-1} e^{-(d-1)D(p' \| 1/2)}$, and $(d - 1)D(p' \| 1/2) \geq (e(C) - \varepsilon)L \ln 2$ for L large. $\square$

Numerically $e(19) = 0.527$, $e(21) = 0.621$, $e(25) = 0.812$, $e(30) = 1.054$; $e(C)$ grows linearly in C with slope $D(1/\log_2 3 \| \frac{1}{2})/\ln 2 = 0.050$.

7.4 The cylinder hypothesis and the bound

Hypothesis H(C, A) (log-log-depth cylinder bound). For all sufficiently large y , with $d = d(y)$, every word $w \in \{O, E\}^d$ beginning with O satisfies

$$\#[w]_y \leq 2^{-(d-1)} \cdot \frac{y}{2} + \frac{y}{(\log y)^A}.$$

Only an upper bound is asked, only at one depth per scale, and only for the $L(y)$ -bad words. Since $2^{-(d-1)}y/2 \asymp y(\log y)^{-C}$, H(C, A) with $A > C$ says that no O -rooted cylinder of depth $d(y)$ exceeds its fair share by more than a relative $O((\log y)^{C-A})$.

Theorem 7.3 (Tao-type bound). Assume H(C, A) with $C \geq 5$ and $A > C + e(C)$. Then for every $\varepsilon > 0$ and all sufficiently large y ,

$$\#\{n \text{ odd} \in (y, 2y] : n \notin R\} \leq \#\{n \text{ odd} \in (y, 2y] : J^t(n) > N_0 \ \forall t \leq d(y)\} \leq \frac{y}{2} \left(\frac{\log 2y}{\log N_0} \right)^{-(e(C)-\varepsilon)}.$$

In words: all but $O(y(\log y)^{-e(C)+\varepsilon})$ odd starts in $(y, 2y]$ enter $[1, N_0]$ within $d(y) = O(\log \log y)$ steps.

Proof. The first inequality is Lemma 7.1 with $d = d(y)$. Every odd start has an O -rooted word; by Lemma 7.2 the O -rooted bad words number at most $2^{d-1} 2^{-(e(C)-\varepsilon)L(y)}$, and by H(C, A) each carries at most $2^{-(d-1)}y/2 + y(\log y)^{-A}$ odd starts. Hence the count is at most

$$\frac{y}{2} 2^{-(e(C)-\varepsilon)L(y)} + 2^{d-1} \frac{y}{(\log y)^A} \leq \frac{y}{2} \left(\frac{\log 2y}{\log N_0} \right)^{-(e(C)-\varepsilon)} + \left(\frac{\log 2y}{\log N_0} \right)^C \frac{y}{(\log y)^A},$$

using $2^{d-1} \leq 2^{CL}$. The second term is $O(y(\log y)^{C-A}) = o(y(\log y)^{-e(C)})$ because $A > C + e(C)$. $\square$

7.5 Contagion closes the loop

Theorem 7.4 (a Tao-type bound with rate implies the conjecture). Suppose that for some $e > 1 - \lambda^{**} = 0.5950\dots$ and all sufficiently large y ,

$$\#\{n \text{ odd}, y < n \leq 2y : n \notin R\} \leq \frac{y}{(\log y)^e}.$$

Then $R = \mathbb{N}$.

Proof. Suppose $F \neq \emptyset$. By Theorem 6.1 every $n \in F$ lies in the E -tree of an odd member of F . For an odd $n_0 \in F$ the level- j set $S_j(n_0)$ of its E -tree has log-mass at most $\prod_{i < j} (1 + n_0^{-2^i})/n_0 \leq 2/n_0$ (Lemma 3.1 gives $\sum_{n \in E(m)} 1/n \leq (m+1)/m^2$), and only levels with $n_0^{2^j} \leq x$, i.e. $j \leq \log_2 \log x$, meet $[1, x]$. Hence

$$\sum_{\substack{n \in F \\ n \leq x}} \frac{1}{n} \leq 2(1 + \log_2 \log x) \sum_{\substack{n_0 \in F \text{ odd} \\ n_0 \leq x}} \frac{1}{n_0} \leq 2(1 + \log_2 \log x) \left(\sum_{k \geq k_0} \frac{2^k (k \ln 2)^{-e}}{2^k} + O(1) \right) \ll (\log x)^{1-e} \log \log x.$$

By Theorem 5.2 the left side is $\geq K(\log x)^\lambda$ for every $\lambda < \lambda^{**}$ and all large x . Choosing $\lambda \in (1 - e, \lambda^{**})$ gives a contradiction. $\square$

Theorem 7.5 (equivalence). Every positive integer reaches 1 if and only if there is $e > 1 - \lambda^{**}$ such that $\#\{n \text{ odd} \in (y, 2y] : n \notin R\} \leq y(\log y)^{-e}$ for all large y .

Proof. If every integer reaches 1 the left side is 0. Conversely Theorem 7.4. $\square$

Corollary 7.6 (the conjecture from a cylinder bound). If $H(C, A)$ holds for some $C \geq 21$ and $A > C + e(C)$, then every positive integer reaches 1.

Proof. Theorem 7.3 gives the hypothesis of Theorem 7.4 with $e = e(C) - \varepsilon \geq e(21) - \varepsilon = 0.621 - \varepsilon > 0.5950$. $\square$

The rate requirement is exactly the complement of the contagion exponent. With λ^{***} of Section 10 it becomes $e > 0.5078$ and $C \geq 19$ suffices.

7.6 The Juggler–Collatz inversion

The mechanism is specific to the Juggler map. Its descent is by powers — one contracting certificate sends n to n^e , $e < 1$ — so the exponent walk has to travel only $\log_2 \log y$ units to bring a start of size y below a fixed N_0 , and the parity word needed is $O(\log \log y)$ letters long, far inside the range where cylinders are large ($y/2^d \gg 1$). For the Collatz map the descent is by constant factors, the walk needs $\Theta(\log y)$ steps, cylinders of that depth have single elements, and Tao’s renewal machinery with a growing target $f(N) \rightarrow \infty$ is forced; moreover thin preimage trees (Krasikov–Lagarias, $x^{0.84}$) mean that even a bounded-target almost-all theorem would not imply the Collatz conjecture. For the Juggler map both obstacles disappear: one-shot Terras counting at depth $O(\log \log y)$ reaches the bounded target, and contagion converts the almost-all statement into the conjecture. What is exact for Collatz — the equidistribution of parity vectors, Terras [4], Everett [5] — is the unproved hypothesis here; what is the wall for Collatz — renewal to a bounded target, and the passage from almost all to all — is free here.

7.7 Constants

With the certified floor $N_0 = 3.5 \cdot 10^8$ and $C = 21$:

y	$L(y)$	$d(y)$	exact fair-coin bad probability	$(\log y)^{-0.6}$	least depth for rate 0.6
10^{20}	1.25	27	0.052	0.100	19
10^{100}	3.55	75	0.014	0.038	56
10^{1000}	6.87	145	0.0028	0.0096	117

The bad probability is the fair-coin probability that a walk of length $d(y)$ never reaches $-L(y)$, by dynamic programming over the walk (unconditioned first letter; conditioning on $u_1 = \log_2 3 - 1$ roughly doubles the finite-depth values and leaves the exponent unchanged); the last column is the least depth at which it drops below $(\log y)^{-0.6}$ — about $17 L(y)$. With the Lean floor $N_0 = 260$ the depths grow by about 38 letters. For comparison, Paper B controls depth 4 (all words) and two words of depth 5, with relative error $y^{-1/96}$ on dyadic blocks; the hypothesis asks for depth $d(y) \rightarrow \infty$ with relative error $o(1)$ — weaker in rate than a power saving, unbounded in depth.

8. Weaker forms of the hypothesis

Theorem 7.3 conditions on the past letter by letter and asks for two-sided control of every cylinder. Neither is needed. This section gives the hierarchy of forms down to the weakest one the concentration argument can use, and records what that form does not need.

8.1 One-sided odd-share control

Hypothesis $H_q(C, A)$. For all sufficiently large y , every $1 \leq t < d(y)$ and every $w \in \{O, E\}^t$,

$$\#\{n \in [w]_y : \text{word}_{t+1}(n) = wO\} \leq q \# [w]_y + \frac{y}{(\log y)^A}.$$

(The depth-0 cylinder is all odd starts, whose next letter is O with share 1; the hypothesis starts at depth 1.)

Theorem 8.1 (biased-split form). Let $q < \log 2 / \log 3 = 0.6309 \dots$, $\mu = 1 - q \log_2 3 > 0$, $C > 1/\mu$, and $e_q(C) = 2(C\mu - 1)^2 / (C(\log_2 3)^2 \ln 2)$. Under $H_q(C, A)$ with $A > C + 1$, for every $\varepsilon > 0$ and all large y ,

$$\#\{n \text{ odd} \in (y, 2y] : n \notin R\} \leq \frac{y}{2} \left(\frac{\log 2y}{\log N_0} \right)^{-(e_q(C) - \varepsilon)}.$$

Hence $e_q(C) > 1 - \lambda^{**}$ implies the conjecture; the least C is 21 at $q = \frac{1}{2}$, 46 at 0.55, 255 at 0.60, 1840 at 0.62.

Proof. Let n be uniform on the odd integers of $(y, 2y]$, $\mathcal{F}_t$ the σ -algebra of the depth- t cylinders, $X_t = \mathbf{1}[\text{word}_{t+1}(n) = \text{word}_t(n)O]$ for $t \geq 1$, and $u_t = (\log_2 3 - 1) + \sum_{1 \leq s < t} (X_s \log_2 3 - 1)$ the exponent walk. Put $\eta_t = (\mathbb{P}(X_t = 1 \mid \mathcal{F}_t) - q)^+$. On a cylinder $[w]$ of depth t , H_q gives $\eta_t \leq y(\log y)^{-A} / \# [w]$, so $\mathbb{E}[\eta_t] \leq 2^{t+1}(\log y)^{-A}$ and $\mathbb{E}[\sum_{1 \leq t < d} \eta_t] \leq 4(\log 2y / \log N_0)^C (\log y)^{-A}$. Since $\mathbb{E}[u_{t+1} - u_t \mid \mathcal{F}_t] \leq -\mu + \eta_t \log_2 3$, the process $M_t = u_t - u_1 - \sum_{1 \leq s < t} \mathbb{E}[u_{s+1} - u_s \mid \mathcal{F}_s]$ is a martingale with increments in an interval of length $\log_2 3$, and $u_d \leq u_1 + M_d - (d-1)\mu + \log_2 3 \sum_{s < d} \eta_s$. Fix $\kappa \in (0, 1)$. If $n \notin R$ then by Lemma 7.1 $u_d > -L$, so either $\log_2 3 \sum_{s < d} \eta_s > \kappa(d-1)\mu$, an event of probability $O((\log y)^{C-A})$ by Markov, or $M_d > a := (1 - \kappa)(d-1)\mu - L - u_1 \geq L((1 - \kappa)C\mu - 1) - \log_2 3$. By Azuma–Hoeffding, $\mathbb{P}(M_d > a) \leq \exp(-2a^2 / ((d-1)(\log_2 3)^2)) \leq 2^{-L(e_q^{(\kappa)}(C) - o(1))}$ with $e_q^{(\kappa)}(C) = 2((1 - \kappa)C\mu - 1)^2 / (C(\log_2 3)^2 \ln 2) \rightarrow e_q(C)$ as $\kappa \rightarrow 0$. $\square$

So no lower bound on odd shares and no vanishing error are needed: if no cylinder of depth below $46 \log_2(\log 2y / \log N_0)$ sends more than 55% of its members to an odd state, every positive integer reaches 1.

8.2 The pressure form

Let $N = y/2$, let $\tau(n) = \min\{t \geq 1 : J^t(n) \leq N_0\}$ be the entrance time into the floor, call n *live at depth t* if $\tau(n) > t$, and write $o_t(n)$ for the number of odd letters among $n, J(n), \dots, J^{t-1}(n)$. By Lemma 7.1, $\tau(n) > t$ implies $u_t(n) > -L(y)$. The stopping is essential: $J(1) = 1$ is odd, so an orbit that has reached 1 has an all- O tail and the unstopped odd count of every start in R grows linearly. For $\theta > 0$ put $a_\theta = \frac{1}{2}(1 + e^\theta)$.

Hypothesis $P_\theta(C)$ (live pressure). For all sufficiently large y , with $d = d(y)$,

$$\frac{1}{N} \sum_{\substack{n \text{ odd} \in (y, 2y] \\ \tau(n) > d}} e^{\theta o_d(n)} \leq a_\theta^d e^{o(d)}.$$

Theorem 8.2 (pressure form). Assume $P_\theta(C)$ with $C \geq 5$ and $\theta = \theta_C = \log(p_C/(1-p_C))$. Then for every $\varepsilon > 0$ and all large y ,

$$\#\{n \text{ odd} \in (y, 2y] : \tau(n) > d(y)\} \leq \frac{y}{2} \left(\frac{\log 2y}{\log N_0} \right)^{-(e(C)-\varepsilon)},$$

the bound of Theorem 7.3; hence $P_{\theta_C}(C)$ with $C \geq 21$ implies the conjecture ($C \geq 19$ with λ^{***}).

Proof. If $\tau(n) > d$ then $u_d > -L$, i.e. $o_d \geq p_C d$. Hence

$$\#\{\tau > d\} \leq \#\{\tau > d, o_d \geq p_C d\} \leq e^{-\theta p_C d} \sum_{\tau(n) > d} e^{\theta o_d(n)} \leq N e^{-\theta p_C d} a_\theta^d e^{o(d)}.$$

At $\theta = \theta_C$, $\theta p_C - \log a_\theta = D(p_C \| \frac{1}{2})$, so the count is $\leq N \exp(-d D(p_C \| \frac{1}{2})(1 - o(1))) \leq N 2^{-(e(C)-\varepsilon)L}$. $\square$

For $1 \leq t < d$ let $\mu_{\theta,t}$ be the probability measure on the starts live at depth t with density proportional to $e^{\theta o_t(n)}$, and let $s_\theta(t) = \mu_{\theta,t}(J^t(n) \text{ odd})$ be the *tilted odd share*: the probability that the next letter is O when odd-heavy live prefixes are up-weighted exponentially.

Hypothesis $M_{\theta,q}(C)$ (no momentum). For all large y , $\sum_{t=1}^{d(y)-1} (s_\theta(t) - q)^+ = o(d(y))$.

Proposition 8.3. $M_{\theta,1/2}(C)$ implies $P_\theta(C)$. More generally $M_{\theta,q}(C)$ with $q < p_C$ gives $\#\{\tau > d\} \leq N \exp(-d D(p_C \| q)(1 - o(1)))$, i.e. the exponent $e_q^{\text{Ch}}(C) = C D(p_C \| q) / \ln 2$, at least the Azuma exponent of Theorem 8.1 (least C with λ^{***} : 19 at $q = \frac{1}{2}$, 41 at 0.55, 214 at 0.60, 1497 at 0.62).

Proof. Dropping the condition $\tau > t+1$ in favour of $\tau > t$ only enlarges the sum, so

$$\sum_{\tau > t+1} e^{\theta o_{t+1}} \leq \sum_{\tau > t} e^{\theta o_{t+1}} = \left(\sum_{\tau > t} e^{\theta o_t} \right) (1 + (e^\theta - 1)s_\theta(t)).$$

With $a_{\theta,q} = 1 + (e^\theta - 1)q$ and $c_\theta = (e^\theta - 1)/a_{\theta,q}$, $1 + (e^\theta - 1)s \leq a_{\theta,q} \exp(c_\theta(s - q)^+)$. Telescoping from $t = 1$ gives $\sum_{\tau > d} e^{\theta o_d} \leq N e^\theta a_{\theta,q}^{d-1} \exp(c_\theta \sum_{t < d} (s_\theta(t) - q)^+)$, which is P_θ at $q = \frac{1}{2}$; the exponential Markov step of Theorem 8.2 with $a_{\theta,q}$ in place of a_θ and $\theta = \log \frac{p_C(1-q)}{q(1-p_C)}$ gives $D(p_C \| q)$. $\square$

8.3 What the weakest form does not need

(a) **Any $o(\log \log y)$ initial depths are free.** A letter contributes a factor at most e^θ to the moment, so bias — however extreme — at any set of depths of size $o(d(y))$ is absorbed by the $e^{o(d)}$. In particular the depth-five split (Paper B's Conjecture 7.3, the K_3 kernel) and every split to any fixed depth, or to depth $s_0(y)$ for any $s_0 = o(\log \log y)$, is irrelevant to the reduction.

(b) **The tower cylinders are nearly free.** A cylinder w of depth t enters P_θ only through $\#[w]e^{\theta o(w)}$, against a fair total $N e^\theta a_\theta^{t-1}$. If the tower O^t splits with odd share β at every level, $\#[O^t] = N \beta^{t-1}$ and its tilted weight relative to the fair total is $(\beta e^\theta / a_\theta)^{t-1}$, which decays geometrically iff $\beta < \frac{1}{2}(1 + e^{-\theta})$, i.e. 0.836 at $\theta_{19} = 0.396$. So a tower biased anywhere below 0.836 contributes a bounded total to $\sum_t (s_\theta(t) - \frac{1}{2})^+$ when the rest of the population is fair; H_q needs the same tower to split below 0.6309, and the conclusion of Theorem 7.3 itself fails only once $\beta > 2^{-e(C)/C} = 0.981$. A single cylinder of depth t matters only if it is over-populated by a factor $(2a_\theta e^{-\theta})^t = (1.67)^t$ relative to its fair share.

(c) **One-sidedness and aggregation.** $M_{\theta,q}$ bounds one weighted average per depth from above; under-populated odd continuations anywhere, and over-populated ones on cylinders of small tilted weight, cost nothing. In Walsh terms, $e^{\theta X_s} = a_\theta + b_\theta(-1)^{J^s(n)}$ with $b_\theta/a_\theta = -\tanh(\theta/2)$, so the unstopped moment is

$$\frac{1}{N} \sum_n e^{\theta o_d(n)} = a_\theta^d \left(1 + \sum_{\emptyset \neq T} (-\tanh \frac{\theta}{2})^{|T|} \frac{W_T}{N} \right), \quad W_T = \sum_n \prod_{s \in T} (-1)^{J^s(n)} :$$

Walsh sums of high order are exponentially down-weighted ($\tanh(0.198) = 0.196$ per letter), whereas the uniform cylinder hypothesis needs all 2^d of them with equal weight.

(d) **Intermediate forms are reparameterizations.** Between $H(C, A)$ and P_θ sit the “almost all cylinders” form (odd share $\leq q$ on all but a family of cylinders of total mass $\leq N(\log y)^{-B}$ per depth, $B > e_q(C)$; the Markov step of Theorem 8.1 absorbs it) and its second-moment form. With $D(w) = \#[wO] - \frac{1}{2}\#[w]$ and the collision count $\mathcal{C}_t = \sum_{|w|=t} \#[w]^2$, Parseval gives $\sum_w D(w)^2 = 2^{-t-2} \sum_{S \subseteq [0,t)} |W_{S \cup \{t\}}|^2 = \frac{1}{2}\mathcal{C}_{t+1} - \frac{1}{4}\mathcal{C}_t$, so the almost-all form follows from the pair-correlation asymptotic $\mathcal{C}_t \leq (N^2/2^{t-1})(1 + (\log y)^{-A'})$, $A' > C + 2$; but by Walsh inversion any such two-sided bound is equivalent, up to the exponent, to polylogarithmic savings on every Walsh sum to depth $d(y)$, i.e. to $H(C, A)$ with another A . They are not weaker in substance; the pressure form is.

(e) **No bounded-depth cylinder statement can close the reduction.** Fix k . The probability measure on parity words that is fair to depth k and all- O afterwards satisfies every cylinder-count statement of depth $\leq k$ exactly, and under it a positive proportion of starts is live for ever. An argument for the Tao-type bound that used only depth- $\leq k$ cylinder statistics of the Juggler map and the walk logic would prove the same bound for that measure. Hence every proof of the bound — in any of the forms above — must use information about the parity words at depth $\rightarrow \infty$; fixed-depth equidistribution at any depth is neither necessary (by (a)) nor sufficient (by (e)) for it.

8.4 Where the wall is

$P_\theta(C)$ says that the exponentially tilted distribution of live parity words of length $d \asymp \log_2 \log y$ has the fair pressure $\log a_\theta$: odd-heavy live words — those with about $e^\theta/(1 + e^\theta) \approx 60\%$ odd letters, the words the tilt selects — are not over-populated at an exponential rate in d . Typical such words contain odd runs of length ≥ 4 , so the nested towers of height ≥ 4 that stop Paper B at depth five sit inside almost every relevant cylinder; but no individual cylinder is required to split fairly, only the tilted average over the $\approx 2^{0.97d}$ typical cylinders at each depth. This is a statement about averaging over cylinders at unbounded depth — a mean over characters, not a sup — of a kind for which analytic number theory sometimes has tools where pointwise bounds fail; we know no such tool for nested-floor parities and record the question, not an approach.

9. One frontier statement

9.1 The free term is the infinite-depth live mass

Recall the free term ψ_F of (6.1). For odd n , $n \in F \iff \lfloor n^{3/2} \rfloor \in F$ (Theorem 6.1), and $n \in F \iff \tau(n) = \infty$ since $[1, N_0] \subseteq R$. Let $\mathbb{P}_x^{\log}$ be the $1/n$ -weighted measure on the odd $n \in (\sqrt{x}, x]$.

Proposition 9.1. $\psi_F(t) = \lim_{d \rightarrow \infty} \mathbb{P}_x^{\log}(\tau(n) > d \mid \text{word}_2(n) = OO)$, a decreasing limit, and under $P_\theta(C)$ — equivalently under any hypothesis of Sections 7–8 —

$$\psi_F(t) \leq 2(1 + o(1)) \left(\frac{t}{2 \log N_0} \right)^{-(e(C) - \varepsilon)}.$$

Proof. The first claim is the definition of ψ_F together with $F \cap \{\text{odd}\} = \bigcap_d \{\tau > d\} \cap \{\text{odd}\}$. For the bound, cover $(\sqrt{x}, x]$ by dyadic blocks $(y, 2y]$; the OO -type odd starts of a block number $\frac{y}{4}(1 + O(y^{-\delta}))$ (the parity of $\lfloor n^{3/2} \rfloor$ over odd n is equidistributed with a power saving — the depth-two case of Paper B, a single exponential sum), the live starts at depth $d(y)$ number at most $\frac{y}{2}(\log 2y / \log N_0)^{-(e(C) - \varepsilon)}$ by Theorem 8.2, and $\{\tau > d\}$ decreases in d . The $1/n$ -weighting only averages the block ratios. $\square$

So the free term of the exact map is the infinite-depth live mass of the depth-two cylinder OO , and the hypotheses of Sections 7–8 bound it. There are not two frontier statements but one quantity — the live mass of odd-heavy parity words at depth $\asymp \log \log x$ — seen once as a limit (exact map) and once at finite depth with a rate (Tao-type bound).

9.2 Exact map = contagion + free term, and the critical exponent

The two halves of (6.1) are the two theorems of Sections 5 and 9. Dropping $\psi \geq 0$ gives the lower recursion $\varphi_F(t) \geq \frac{1}{2}\varphi_F(t/2) + \frac{1}{4}\varphi_F^{\text{fib}}(3t/4)$, which in log-mass form $g = (t/2)\varphi$ is $g(t) \geq g(t/2) + \frac{1}{3}g^{\text{fib}}(3t/4)$: the ideal

depth-two contagion recursion of Section 5 with perfect fibers (the proved constants give $\lambda^{**} = 0.4050$, the ideal ones $\lambda_{\text{ideal}} = 0.4927$). Dropping $\psi \leq 1$ instead gives the upper recursion for the failure density. Its homogeneous solutions $\varphi = t^{-e}$ satisfy $\frac{1}{2}2^e + c(4/3)^e = 1$ with $c = \frac{1}{4}$ for ideal fibers and $c = \frac{2}{7}$ with the two-sided sweep constant $\frac{4}{7}$ of Lemma 4.2:

$$e_{\text{crit}} = 1 - \lambda_{\text{ideal}} = 0.5073 \quad (\text{ideal}), \quad e_{\text{crit}} = 0.4354 \quad (\text{sweep}).$$

The rate threshold $e > 1 - \lambda^{**}$ of Theorem 7.4 (or $1 - \lambda^{***} = 0.5078$ with Section 10) is therefore the homogeneous decay exponent of the exact map up to the fiber-constant gap: contagion's growth rate and the exact map's decay rate are the same number because they are the same recursion.

Corollary 9.2 (the exact map cannot replace contagion). Suppose $\psi_F(t) \leq t^{-e}$ with e as large as one likes. The upper recursion then gives $\varphi_F(t) \ll t^{-\min(e, e_{\text{crit}})}$ and nothing more: with honest constants $\varphi_F \ll t^{-0.435}$, ideally $\varphi_F \ll t^{-0.507}$, never $\varphi_F \equiv 0$. Contagion gives $\varphi_F \gg t^{-(1-\lambda)}$ if $F \neq \emptyset$. The two bounds do not cross. What crosses contagion is the direct inclusion $F \cap (y, 2y] \subseteq \{\tau > d(y)\}$ applied to all starts — Theorem 7.4 — not the recursion. Conversely, if $F \neq \emptyset$ and ψ_F decayed faster than the critical rate, the sandwich would force $\varphi_F(t) \asymp t^{-0.507}$: a nonempty failure set with a small free term would have log-count exactly of the order $(\log x)^{0.4927}$.

9.3 The depth-uniformity budget

Section 8.3(e) says that a proof of the Tao-type bound must use information at depth $\rightarrow \infty$. The form of Theorem 7.3 says how uniform that information must be. Suppose an analytic method proves cylinder equidistribution at depth d with a power saving, $\#[w]_y = 2^{-(d-1)\frac{y}{2}} + O(y^{1-\delta_d})$, where the saving exponent loses a factor 2^c per depth: $\delta_d = \delta_0 2^{-cd}$ (Weyl differencing loses $c \geq 1$; Paper B's chain $\frac{1}{24} \rightarrow \frac{1}{96}$ from depth three to four is $c = 2$). Theorem 7.3 needs the total error $2^d y^{1-\delta_d}$ to be $o(y(\log y)^{-e(C)})$ at $d = C \log_2 \log y$, i.e. $\delta_d \log y \geq (C + e(C)) \log \log y$, i.e. $(\log y)^{1-cC} \gtrsim \log \log y$:

$$cC < 1, \quad \text{so the per-depth loss factor must be below } 2^{1/C} = 1.037 \ (C = 19), \ 1.017 \ (C = 41).$$

Any method losing more than about 4% of its saving exponent per depth is structurally excluded from the Tao depth, whatever it does at depth five; Weyl differencing, at $c \geq 1$, reaches at most depth $\approx \log_2 \log y$ — one unit of $L(y)$, enough for the all- E word to reach the floor, not for the Chernoff tail, which needs $C \geq 5$ units. A polylogarithmic relative saving at every depth, as in $H(C, A)$, would suffice; what is excluded is power-saving machinery whose exponent decays exponentially in the depth.

10. A conditional strengthening: localizing Paper B

Everything so far is unconditional (Theorems 1, 2, 4, 5) or conditional only on the stated hypotheses (Theorem 3). This section depends in addition on Paper B [12], a working draft whose analytic estimates have not been independently refereed; its results are labelled accordingly and no statement of Sections 1–9 relies on them.

The contagion recursion (5.2) uses the productions E and OE . A deeper contracting word w enters it with coefficient P_w/e_w at the argument $e_w t$, where P_w is the fair probability of the word and x^{e_w} the scale of its landing point, *provided* the cylinder $[w]$ is equidistributed on the preimage interval of each landing point — an interval of length x^{1-e_w} at scale x , sub-dyadic for every $e_w > 0$. Paper B proves its equidistribution theorems on dyadic blocks. We checked that the Weyl-differencing proof of its Theorems 4.4 and 4.7 (the triple parity discrepancy of $n^{3/2}$, $\lfloor n^{3/2} \rfloor^{3/2}$, $\lfloor \lfloor n^{3/2} \rfloor^{3/2} \rfloor^{1/2}$ over odd n) localizes.

Proposition 10.1 (conditional on Paper B; localized triple parity discrepancy). With $\psi(y) = (-1)^{\lfloor y \rfloor}$, $m = \lfloor n^{3/2} \rfloor$, $v = \lfloor m^{3/2} \rfloor$, $\psi_1 = \psi(n^{3/2})$, $\psi_2 = \psi(m^{3/2})$, $\psi_3 = \psi(v^{1/2})$: for every $\varepsilon > 0$ there is P_0 such that for all $P \geq P_0$, every interval $I \subseteq (P, 2P]$ of length $Y \geq P^{1/2}$, and every $(a, b, c) \in \{0, 1\}^3 \setminus \{0\}$,

$$\left| \sum_{n \in I, n \text{ odd}} \psi_1^a \psi_2^b \psi_3^c \right| \leq Y P^{-1/24+\varepsilon}.$$

Proof sketch. Run the seven steps of Paper B's proof (wave expansion, exact linearization, Weyl differencing with $H = P^{1/12}$, the differenced phase, cells, the second-derivative test per cell, assembly) over I instead of $(P, 2P]$, with the same truncations. Every displayed bound is of one of two kinds: a bound proportional to the number of summands, which becomes proportional to Y ; or a bound independent of the number of summands, which is unchanged and is at most $P^{7/16}$ (the two end cells of the cell decomposition, $P^{3/8}$; the pure passenger term, $P^{7/16}$; majorant tails, $P^{1/4}$). For $Y \geq P^{1/2}$ the latter fall below the main saving $YP^{-1/24}$. The modification list with its constants is recorded in the laboratory note on fate contagion, Section 7. $\square$

Proposition 10.2 (conditional; the OOOEE production on even blocks). For $m' \geq 2$ let $I(m') = [m'^{32/9}, (m' + 1)^{32/9})$, of length $\asymp P^{23/32}$ at scale $P = m'^{32/9}$, and $\mathcal{O}(m') = \{n \text{ odd} \in I(m') : \text{word}_5(n) = \text{OOEEE}, J^5(n) = m'\}$. Every $n \in \mathcal{O}(m')$ has $J^4(n) \in E(m')$, and

$$|\mathcal{O}(m')| = \frac{1}{16} \# \{n \text{ odd} \in I(m')\} + O_\varepsilon(|I(m')| m'^{-4/27+\varepsilon}).$$

Consequently, for a backward-closed A and $m' \in A$, $\mathcal{O}(m') \subseteq A$ with log-mass at least $\frac{1}{9m'}(1 - O(m'^{-4/27+\varepsilon}))$.

Proof sketch. Along OOOEE the states are n (odd), m (odd), v (even), $\lfloor v^{1/2} \rfloor$ (even), $\lfloor v^{1/4} \rfloor$ (even), and $J^5(n) = \lfloor \sqrt{J^4(n)} \rfloor$; $J^5(n) = m'$ forces $J^4(n) \in E(m')$. The transparent nesting $\lfloor v^{1/4} \rfloor = \lfloor n^{9/16} \rfloor$ holds off an exceptional set of n with $\{n^{9/16}\} < n^{-15/16}$, of size $\ll YP^{-1/24} + P^{7/16}$ by Erdős–Turán with Kusmin–Landau; off it, $n \in I(m')$ forces $J^5(n) = m'$. The indicator of OOOEE on odd n is $\frac{1}{16}(1 - \psi_1)(1 + \psi_2)(1 + \psi_3)(1 + \psi_4)$ with $\psi_4 = \psi(v^{1/4})$; the fifteen sign sums without ψ_4 are Proposition 10.1 with $Y = |I(m')| \geq P^{1/2}$, and those with ψ_4 are handled by replacing $v^{1/4}$ with $n^{9/16}$ (cost $\ll YP^{-15/16}$) and Vaaler-expanding $\psi(n^{9/16})$: its modes have curvature $\ll P^{-23/16}$, far below every curvature retained in the cell analysis, so they ride along, and their pure sums are $\ll P^{7/16}/\ell$ by Kusmin–Landau. Every error is $\ll YP^{-1/24+\varepsilon} = Ym'^{-4/27+\varepsilon}$. The log-mass bound follows from $1/n \geq (m' + 1)^{-32/9}$ and $\frac{1}{16} \cdot \frac{1}{2} \cdot \frac{32}{9} m'^{23/9} = \frac{1}{9} m'^{23/9}$. $\square$

Theorem 10.3 (conditional; improved exponent). Under Propositions 10.1–10.2, Theorem 5.2 holds for every $\lambda < \lambda^{***} = 0.4922\dots$, the root of

$$2^{-\lambda} + \frac{5}{21} \left(\frac{3}{8}\right)^\lambda + \frac{2}{21} \left(\frac{3}{4}\right)^\lambda + \frac{1}{9} \left(\frac{9}{32}\right)^\lambda = 1.$$

Proof. Add to the three families of Section 5.1 a fourth: for $m' \in A \cap (x^{9/64}, x^{9/32} - 1]$, the OOOEE-starts of $I(m') \subseteq (\sqrt{x}, x]$. They are odd with odd $\lfloor n^{3/2} \rfloor$, hence disjoint from the even members and from the OE-type members of the first three families, and distinct m' give disjoint $I(m')$. By Proposition 10.2 their log-mass is at least $\frac{1}{9}(1 - \varepsilon_7(x))(g_A(9t/32) - 2x^{-9/32})$, and (5.2) gains the term $(\frac{1}{9} - \varepsilon_7)g_A(9t/32)$. The induction of Section 5.4 runs verbatim with the new root. $\square$

Downstream, every constant of Sections 7–9 improves: the rate threshold of Theorems 7.4–7.5 becomes $e > 1 - \lambda^{***} = 0.5078$; the least depth constant is $C = 19$ ($e(19) = 0.527$); in the one-sided form $C(0.5) = 19$, $C(0.55) = 41$, $C(0.60) = 223$, $C(0.62) = 1587$; in the pressure form 19, 41, 214, 1497. The exponent 0.4922 is within 0.0005 of the depth-two ceiling 0.4927, reached by a different route. Further gains from the words OOOEEE and OOEOEE on even blocks would need Paper B's kernel theorem (its Theorem 5.3, a double Weyl differencing) on intervals of length $P^{23/32}$; the proof has the same scaling architecture, and the localization is plausible, but its forty-odd displayed estimates have not been re-derived, and we do not use it. It would give $\lambda = 0.5561$ and $C = 18$. None of this touches the free term ψ_F : Propositions 10.1–10.2 concern consecutive odd starts in an interval and say nothing about the odd images S_{odd} .

11. Observations

The three censuses below are numerical checks. They prove nothing; they show that the hypotheses of Section 8 hold in aggregate over the accessible range and that the constants of Section 4 are conservative. All are computed by exact integer arithmetic (`research.juggler_sequence.fate_contagion`, `research.juggler_sequence.tao_reduction`; artifacts in Appendix B).

11.1 Fibers, blocks, and the certified closure

Fibers. Over $3375 \leq m < 10^5$ and spot ranges at 10^6 and 10^7 , the mean of G_m/H_m is 0.5000, 0.5004, 0.5027; the minimum on good fibers is 0.328 at $m = 1003635$ ($\alpha_m \approx \frac{1}{3}$, the three-cluster case); no good fiber falls below the sweep bound $\frac{1}{7}$. The truth on good fibers is about $\frac{1}{3}$; the proof gives $\frac{1}{7}$.

Blocks. On $m' \in [20, 60] \cup [200, 230] \cup [1000, 1010] \cup \{3000, 5000\}$ the deviation of $|U(m')|$ from $\frac{1}{4}\#\{n \text{ odd in } I(m')\}$ is at most $0.72\sqrt{|I(m')_{\text{odd}}|}$, square-root scale, and at most $0.078 m'^{11/9} \log m'$.

Certified closure. Let $\mathcal{C}$ be the closure of the Lean-verified seed $[1, 260]$ under the productions E and OE of Section 3; every member reaches 1. Computed exactly up to 10^9 , $\mathcal{C}$ has density between 0.45 and 0.49 on every dyadic block $(2^k, 2^{k+1}]$ with $16 \leq k \leq 29$, against 0.25 for the E -only closure. At $x = 10^9$ the log-mass of $\mathcal{C} \cap (\sqrt{x}, x]$ is $5.333 = 3.785$ (even) $+ 1.547$ (odd), and the realized recursion coefficients are 0.9998 for E (theory 1) and 0.3332 for OE against the source mass $\ell_{\mathcal{C}}(x^{3/8}, x^{3/4})$ (heuristic $\frac{1}{3}$; proved $\frac{1}{3}$ on E -blocks and $\frac{2}{21}$ elsewhere). The first-letter decomposition of $\mathcal{C}$ illustrates the free term of (6.1): its OE -type share of members equals the fair share to three digits (0.7288 against 0.7305 at $x = 10^6$), while its OO -type share is 0 against a fair 0.30 — the free term is exactly the omitted production.

11.2 The Tao census: survival above the certified floor

Because every $n \leq N_0 = 3.5 \cdot 10^8$ reaches 1, the statistic “ $J^t(n) \leq N_0$ for some $t \leq d$ ” is a finite computation on exact integers, and its complement is the bad set of Theorem 7.3. For random odd starts in $(y, 2y]$, the fraction of orbits still above N_0 after d steps, against the odd-start fair-coin survival $\Pr[u_t > -L(y) \ \forall t \leq d \mid u_1 = \log_2 3 - 1]$ (exact dynamic programming):

y	$L(y)$	$d = 10$	$d = 20$	$d = 30$	$d = 40$	ratio
10^{12}	0.526	0.250/0.250	0.082/0.084	0.043/0.043	0.0215/0.0219	0.98–1.01
10^{15}	0.841	0.291/0.309	0.100/0.106	0.050/0.055	0.026/0.028	0.91–0.95
10^{20}	1.249	0.360/0.383	0.148/0.153	0.068/0.068	0.038/0.038	0.94–1.02
10^{30}	1.826	0.442/0.440	0.207/0.206	0.107/0.103	0.053/0.051	1.00–1.07
10^{50}	2.558	0.618/0.617	0.267/0.264	0.149/0.143	0.087/0.084	1.00–1.05

Entries are empirical/fair; samples 40000 at 10^{12} , 10^{15} , 10^{20} and 20000 at 10^{30} , 10^{50} ; the ratio column is the range of empirical over fair for $d \geq 10$. The only systematic deviation is at depth ≤ 8 for $L(y)$ just below an integer (10^{15} , 10^{20}), where the floors in $J^t(n) \leq n^{2^{u_t}}$ make real orbits descend one step earlier than the continuous threshold — in the favourable direction. The aggregate odd share along surviving prefixes is $\frac{1}{2}$. This is evidence about the aggregate over cylinders; the hypotheses of Section 8 are about every cylinder, or about the tilted aggregate, which no sample can establish.

11.3 The pressure census: no momentum to depth 40

On the live starts of 40000 exact orbits at each of $y = 10^{12}, 10^{20}, 10^{30}, 10^{50}$, the tilted odd share $s_{\theta}(t)$ of Section 8.2 was computed for $t = 1, \dots, 40$ and $\theta \in \{0.396, 0.6\}$ ($0.396 = \theta_{19}$). Live populations at depth 40: 858, 1534, 2069, 3408. Results: $s_{\theta}(t) \in [0.44, 0.56]$ at every depth and scale, within the binomial noise of the effective tilted sample; the cumulative excess $\sum_{t < 40} (s_{\theta}(t) - \frac{1}{2})^+$ is 0.14–0.19 at $\theta = 0.396$ and 0.19–0.26 at $\theta = 0.6$, which is the expected positive-part noise (about $0.4\sigma_t$ per depth); the live moment $\frac{1}{N} \sum_{\tau > d} e^{\theta a_{\tau}}$ divided by its fair-coin value is between 0.95 and 1.08 for $d = 10, 20, 30, 40$. No momentum is detectable to depth 40 at any scale up to 10^{50} . The hypothesis $M_{\theta, q}(C)$ is about $d(y) \rightarrow \infty$.

12. Limitations, and what a proof must do

What is proved. Theorem 1 and its corollaries, Theorem 4, the conditional Theorems 3 and 5, and the equivalence Theorem 2 — all unconditional in the sense that they depend only on the stated hypotheses and on the two exact productions. Section 10 depends on Paper B.

What is not proved. Termination. No cycle is excluded, no divergent orbit is excluded, and no estimate in this paper or in Papers A and B controls the orbits of almost all starts down to a fixed bound. The censuses are observations.

What a proof must do. Every route available to us — cycle finance and the walk charge (Paper A), kernel equidistribution (Paper B), contagion (Section 5), the exact map (Section 6) — bounds a side of (6.1) that does not contain ψ_F . Sections 8–9 reduce the remaining problem to one statement:

The frontier statement. The number of odd starts in $(y, 2y]$ whose orbit is still above the certified floor N_0 after $C \log_2 \log y$ steps is at most $y(\log y)^{-e}$ for some $e > 1 - \lambda^{**}$ (0.595; 0.508 with Section 10).

Its weakest sufficient condition on the concentration route is the no-momentum hypothesis $M_{\theta,q}(C)$: the tilted odd share of live starts exceeds $q < \log 2 / \log 3$ by $o(\log \log y)$ in total over depths. That condition is one-sided, aggregated over cylinders, insensitive to any $o(\log \log y)$ initial depths and to towers biased below 0.836; it cannot be reached from cylinder statistics of bounded depth; and an analytic method that loses more than a factor $2^{1/C}$ of its saving exponent per depth cannot reach the depth it requires. The free term of the exact map is its *OO*-restriction at infinite depth, and contagion is the mechanism that turns it into the conjecture. What is needed, in the end, is a mean-over-cylinders statement about the parities of the nested floor powers $[\dots \lfloor \lfloor n^{3/2} \rfloor^{3/2} \rfloor \dots]$ at depth of order $\log \log n$ — the Terras step of the Juggler map.

Appendix A. Lean names

All in `formal/Problems/Juggler/FateContagion.lean` unless noted; the module compiles with `lake build Problems.Juggler` without `sorry`.

Statement	Lean
backward closure	<code>BackwardClosed</code> , <code>backwardClosed_iterate</code>
Lemma 2.1, fate classes closed	<code>reachesOne_backwardClosed</code> , <code>not_reachesOne_backwardClosed</code> , <code>ancestor_backwardClosed</code> , <code>escapes_backwardClosed</code> , <code>reachesOne_floorPower</code> , <code>escapes_floorPower</code>
Lemma 2.1, trichotomy	<code>Periodic</code> , <code>periodic_of_repeat</code> , <code>fate_trichotomy</code> (from <code>cycles_or_escapes</code>)
Lemma 2.1, exclusion	<code>reachesOne_bounded</code> , <code>reachesOne_not_escapes</code> , <code>cycle_basin_bounded</code> , <code>cycle_basin_not_escapes</code> , <code>reachesOne_not_cycle_basin</code> , <code>periodic_iterate_mod</code>
Lemma 3.1 (even block)	<code>floorPower_even_block</code> , <code>even_block_mem</code> , <code>even_block_card</code>
Lemma 3.2 (<i>OE</i> fiber)	<code>sqrt_sqrt_eq_iff</code> , <code>floorPower_oe_fiber</code> , <code>oe_fiber_mem</code> , <code>oe_fiber_disjoint</code>
Theorem 6.1 (odd generation)	<code>ForwardClosed</code> , <code>reachesOne_forwardClosed</code> , <code>not_reachesOne_forwardClosed</code> , <code>escapes_forwardClosed</code> , <code>mem_iff_floorPower_mem</code> , <code>exists_odd_ancestor</code> , <code>exists_odd_ancestor_ge_three</code> , <code>nonempty_iff_odd_image_mem</code> , <code>odd_mem_iff</code>
Lemma 7.1 (envelope descent)	<code>iterate_le_of_envelope</code> , <code>mem_of_envelope_floor</code> , <code>reachesOne_of_itinerary_envelope</code> ; <code>power</code> <code>envelope_power_bound_word</code> (Paper A layer)
Lean floor $N_0 = 260$	<code>reachesOne_of_lt_two_hundred_sixty_one</code>

Statement	Lean
Lemmas 4.1–4.3, Proposition 4.4, Theorem 5.2, Sections 7–9	human proofs

Appendix B. Constants and artifacts

Exponents. Roots of $2^{-\lambda} + \sum_i c_i e_i^\lambda = 1$:

recursion	terms (c_i, e_i)	root
elementary sweep only	$(\frac{2}{21}, \frac{3}{4})$	0.1385
block average only (λ^*)	$(\frac{1}{3}, \frac{3}{8})$	0.3774
block average + sweep (λ^{**})	$(\frac{5}{21}, \frac{3}{8}), (\frac{2}{21}, \frac{3}{4})$	0.4051
+ <i>O O E E E</i> on even blocks (λ^{***} , Section 10)	$\dots, (\frac{1}{9}, \frac{9}{32})$	0.4923
depth-two ideal	$(\frac{1}{3}, \frac{3}{4})$	0.4927
+ <i>O O O E E E</i> , <i>O O E O E E</i> (not proved)	$\dots, (\frac{2}{27}, \frac{27}{64})$	0.5561

Depth constants. Chernoff exponent $e(C)$: 0.480 (18), 0.527 (19), 0.574 (20), 0.621 (21), 0.812 (25), 1.054 (30). Least C for the rate threshold $1 - \lambda^{**} = 0.5950$ / $1 - \lambda^{***} = 0.5078$: fair Chernoff 21/19; one-sided Azuma at $q = 0.5, 0.55, 0.60, 0.62$: 21, 46, 255, 1840 / 19, 41, 223, 1587; pressure (biased Chernoff) at the same q : 21, 46, 245, 1735 / 19, 41, 214, 1497. Depth $d(y) = \lceil 21L(y) \rceil$ with $N_0 = 3.5 \cdot 10^8$: 27 at 10^{20} , 75 at 10^{100} , 145 at 10^{1000} , 215 at 10^{10000} .

Artifacts (repository `sneakyweasel/balanced_ternary`, commit `b255996b`; SHA-256):

file	hash
<code>data/research/juggler/fate_contagion/summary.js</code>	<code>60824ce5d5c7ba2ad745f9f912aaf883f7ce7c150b3de127eab22ca187</code>
<code>data/research/juggler/tao_reduction/summary.js</code>	<code>0a7c3e396d02644d27d676e1e9a20c12d35fa7e6eb8df342ae74362d4f</code>
<code>formal/Problems/Juggler/FateContagion.lean</code>	<code>cac6a00884346fcc98a03603bf919e03a681f8f66b8b55f3cff9d336e</code>
<code>src/research/juggler_sequence/fate_contagion.py</code>	<code>f1e9c899c5178b971a8bbfba8d46ac033b1426f1a19c8a18ead2efcda</code>
<code>src/research/juggler_sequence/tao_reduction.py</code>	<code>98294d543da6f7850627fef773764c2ef1c0d8332ae639bae2d1b5019</code>

The censuses of Section 11 use fixed random seeds and are reproduced by `python -m research.juggler_sequence.fate_con` and `python -m research.juggler_sequence.tao_reduction`.

Acknowledgments

The exact combinatorial layer was developed with machine assistance and checked in Lean 4; the analytic counting, the reduction, and the prose are the author’s responsibility. The laboratory notes from which this paper is distilled record every intermediate step, including the ideas that failed.

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